

CPPS2026

TRUE NORTH POWER
PLATFORM EXPERIENCE



Designing Beautiful Apps for Functional Makers

A Masterclass in Fusing
Power Apps Capabilities



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AGENDA

Introductions / Housekeeping (10 minutes)

Module 1: Overview of Key Topics (20 minutes)

Lab 0: Setup lab environment (15 minutes)

Module 2: Model-Driven Apps (30 minutes)

Lab 1: Create a Model-Driven App (60 minutes)

Module 3: Working with Custom Pages (60 minutes)

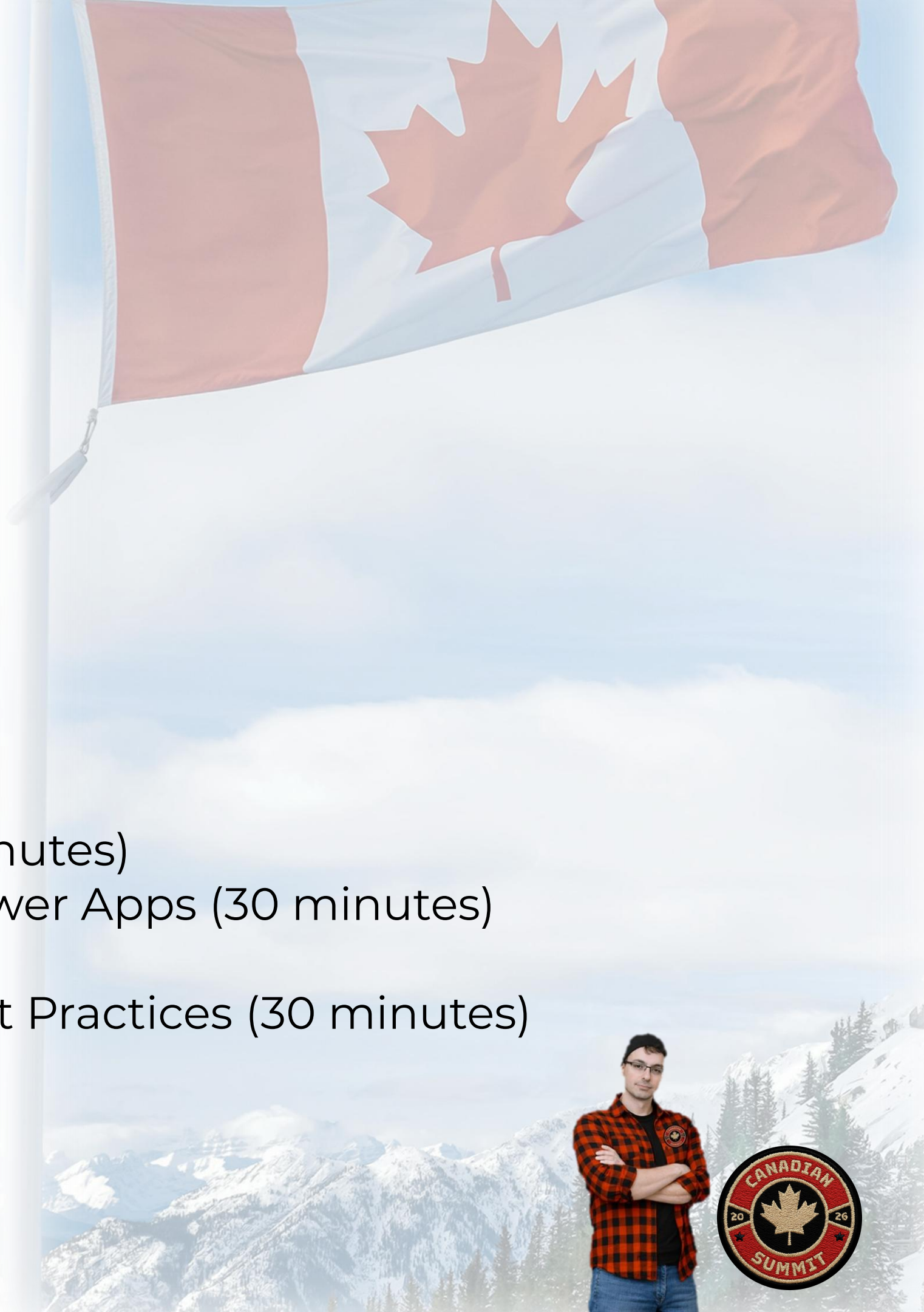
Lab 2: Creating and working with custom pages (60 minutes)

Lab 3: Integrating custom pages with model-driven Power Apps (30 minutes)

Module 4: Going Further - Advanced Concepts and Best Practices (30 minutes)

Lab 4: Test and deploy your solution (30 minutes)

#NoFuglyApps!





Head of Power Platform 🚀 **Point Taken**
Business Applications **MVP**



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CATHRINE BRUVOLD





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📍 United Kingdom

🐙 github.com/JJGriffin

🌐 linkedin.com/in/joejgriffin/



JOE GRIFFIN





But it's not all about us...

Who are you and why are you here?



Please tell us about yourself!

<https://www.menti.com/alsc95eoooh4b>

HOUSEKEEPING



Break times:

- 09:30-09:45 (Timmies Run #1 ☕ 🍁)
- 12:00-13:00 (Lunch)
- 14:30-14:45 (Timmies Run #2 ☕ 🍁)

Wi-Fi Details:

- **SSID:** MSFT-GUEST
- **Attendee Code:** msevent183zw

Today's workshop has **four** hands-on lab exercises:

- You should have already received an email with the instructions.
 - Did you complete the form? 😬 ----->
- If completing in your own environments, Lab 0 should have already been completed.
- Don't have an environment? Don't worry; we will provide access and give you time to complete the lab.
- Links on the next slide to access everything – lab instructions, environment credentials and this slide deck 🙌

Designing Beautiful Apps for
Functional Makers - Lab
Instructions





CONTENT AND LINKS



Lab Instructions

<https://bit.ly/BeautifulPowerAppsLabs>



Environment Credentials

<https://bit.ly/BeautifulPowerApps-Credentials>



Presentation

<https://bit.ly/BeautifulPowerApps-Presentation>

Module 1: Overview of Key Topics





A Powerful Platform

A suite of tools meant to improve and simplify business processes



Copilot Studio

Task-oriented agents and natural language conversation about data.



AI Builder

Build robust, dynamic automations. Invocation of AI models in business processes



Power Apps

Custom applications for solving business needs. UI for exploring organisational data



Power Automate

Automation of workflows and manual processes



Power Pages

Portal and web interface for accessing Dataverse externally



Dataverse

Data storage, security layer and relational database.



The platform is more than just *apps, AI and flows*

You need data, security, connectivity, control and logic...



Power Fx

The language in Power Platform. Formulas to execute logic. Canvas apps, Copilot Studio, Dataverse, Power Pages, Power Automate.



Connectors

Access external data sources and services. Custom connectors (security, reusability, custom API).



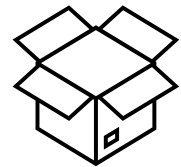
Managed Environments

Overview, control, usage insights, governance, security features and Power Platform Pipelines



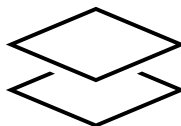
Structure, *security* and *deployment*

How your work is organised, protected and promoted



Solutions

Packaging and deployment containers. Move components between environments (dev, test, prod) with ALM best practices.



Environments

Isolated spaces for building and running apps.
Separate dev, test and production to control change and access.



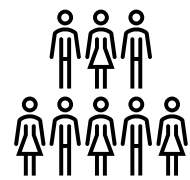
Security Model

Roles, business units and Dataverse table-level permissions. Control who sees and edits what.



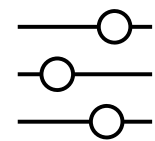
People, *process* and *governance*

Who builds, who controls and how the organisation scales



Citizen Developer / Pro Developer

Low-code for business users, code-first extensibility for developers. PCF controls, plug-ins and Azure Functions bridge the gap.



Governance & CoE

Centre of Excellence toolkit, DLP policies and admin controls. Manage adoption at scale without limiting innovation.



Ahh, the app synergies

Build solid solutions using different application types

Canvas Apps

Task focused. Mobile Apps. Drag and drop. Custom UI. Connectors to access external data.

Model-driven apps

Dataverse driven. Structured data in views and forms. Business processes. JavaScript.

Custom Pages

Custom UI and connectors in MDA. Canvas apps capabilities. Modern approach to embedding UI in MDA

Generative pages/Vibe

React. Prompt and natural language. AI-driven. Modern UI in MDA.

Code Apps

Code first approach. Standalone app. Full control. React - TypeScript



Choosing the **right** tool for your scenario

A quick guide

Need custom quickly → **Generative Pages/Vibe**

Data driven → **Model-driven apps**

Multiple data sources → **Canvas**

MDA functional gaps → **Custom page**

Pro-dev capabilities → **Code Apps**



Choosing the right app type

When to use each — and when to combine them

▶ MODULE 1 - 20m

LAB 0

MODULE 2

LAB 1

MODULE 3

Model-driven apps

- Dataverse is your primary data source
- Standardised forms, views and dashboards
- Complex data model with relationships
- Security roles control access per table



Canvas apps

- Highly customised UI and pixel-perfect layout
- Multiple or non-Dataverse data sources
- Mobile-first or task-specific experiences
- Simple data models, focused workflows



Custom pages

- Canvas-style UI inside a model-driven app
- Replace dialogs, wizards or complex forms
- Combine Dataverse context with custom layout
- Best of both worlds — embedded flexibility



Code-first apps

- Requirements exceed low-code capabilities
- Full developer control (React, PCF, Azure)
- Complex integrations or real-time processing
- Dedicated dev team for build and maintain



+

Combine them

MDA + Custom Pages for flexible forms · Canvas embedded in MDA for task UIs · PCF controls inside any app



Generative pages

Describe a page → AI writes the code → you refine it

What are they?

AI-generated pages inside **model-driven apps**. Describe what you want in plain language, attach a sketch or wireframe, select your Dataverse tables — and an AI agent generates **real React & TypeScript code**.

Why it matters

- Concept to functional page in **minutes, not weeks**
- No drag-and-drop limitations — full creative freedom
- Powered by GPT-5 for high-quality code generation
- Full code transparency — view, edit & extend the React code
- Solution-aware — managed ALM across environments
- Now **generally available** (since November 2025)

✓ Generally available

Available in model-driven apps • Dataverse tables only • Up to 6 tables per page

Think of it as a **reimagined Custom Page** — no Power Fx needed, no drag-and-drop limits, just describe and go

How it works

- 1 Describe**
Write a natural language prompt
- 2 Connect**
Select Dataverse tables + attach sketch
- 3 Generate**
AI writes React & TypeScript code
- 4 Refine**
Iterate via conversation or edit code

▶ **MODULE 1 - 20m**

LAB 0

MODULE 2

LAB 1

MODULE 3



... And know when to **avoid** the app type

A quick guide

Generative Pages → Complex workflows and Multiple data sources, **not just Dataverse**

Model-driven apps → Dataverse is not the data source and a need for highly **customised UI**

Canvas Apps → There are heavy **data operations** and complex data model

Custom page → Need for a **standalone app** and there are heavy data operations

Code Apps → No developer capabilities, which makes a **low code** project too complex



Power Apps is **not dead...** yet

What's new and exciting heading into 2026

▶ **MODULE 1**

LAB 0

MODULE 2

LAB 1

MODULE 3

01

Vibe Coding

Build apps by describing them in plain language at vibe.powerapps.com — AI generates React code, data models and UI automatically

02

Code Apps (GA)

Deploy React, Vue or any web framework directly to Power Platform with full ALM, governance and enterprise guardrails

03

MCP Server & Agent Feed

AI agents fill forms, create records and hand off to humans for review — with a shared supervision workspace

04

M365 Copilot in Apps

Copilot chat embedded in model-driven apps — ask questions, summarise data and take action without leaving the app

05

Plan Designer

Describe a business problem and a team of AI agents generates requirements, data model, app and Power BI reports

06

Modern Controls & Cards

New responsive Card control, Fluent UI dialogs, theme copy-paste across apps and refreshed MDA navigation

The platform is evolving from **low-code** to **AI-first** — where humans and agents build, run and manage apps together



What are we building today

Fusing Power Apps capabilities and creating a beautiful app.

Scenario

Coho Winery needs to access and work with purchase orders from Power Platform. They have data stored in an ERP database and they have relevant documents in OneDrive.

The task

Develop an end-to-end solution that integrates data from Coho Winery's ERP database into a model-driven app to make purchase order information more accessible for the users.

You will:

- Utilise a custom API for connecting to ERP data
- Create a model-driven app that has key information about purchase orders
- Develop a custom UI in two different custom pages to serve as a **landing page** and as a **side pane** to display purchase order PDFs and embed them in the MDA.



Todays Build

Extending model-driven apps with pixel-perfect custom UI

► **MODULE 1 - 20m**

LAB 0

MODULE 2

LAB 1

MODULE 3

What is a Custom Page?

A canvas-based page that runs inside a model-driven app. It uses Power Fx and the canvas designer, giving you full control over layout, styling and behaviour — all within your MDA.

Why use them?

Pixel-perfect UI



Full canvas designer control over layout, colours and components

Embedded experiences



Display as full page, side pane or dialog — all within the MDA

Power Fx + connectors



Access 1,000+ connectors and custom APIs alongside Dataverse

Solution-aware



Managed ALM — deploy across environments with your MDA solution

In our workshop

Coho Winery — Purchase Order app

Model-driven app

Custom Page

Landing Page

Dashboard-style overview with KPIs, charts and navigation tiles

Custom Page

Side Pane

Displays purchase order PDFs alongside the form — no new tabs

Model-driven forms

Tables, views, forms, dashboards — Dataverse-driven

Dataverse · Custom API · OneDrive · ERP connector

Custom pages bridge the gap between canvas flexibility and model-driven structure — one app, best of both worlds



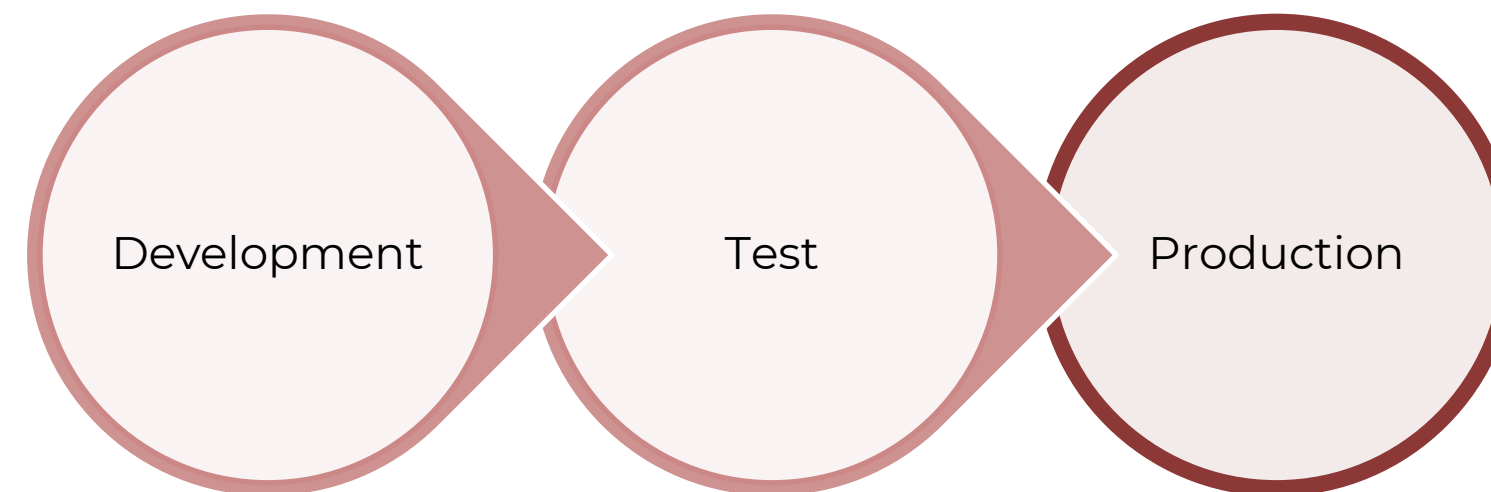
DEMO: What we will build today



First, we need a solid foundation

The importance of Solutions and Application Lifecycle Management (ALM)

- Solutions are containers for Power Platform components 📦
- Transport customizations from one environment to another
- Part of an overall strategy for application lifecycle management (ALM)
- Unmanaged versus managed solutions



So easy, so no excuse



Creating a solution

- **Display name** used in Maker portal
- **Name** is the schema name and cannot be changed and cannot have spaces
- **Publisher** defines the prefix for new components
- **Version** number to track updates

New solution



Display name *

Name *

Publisher *

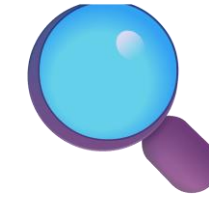
+ New publisher

Version *

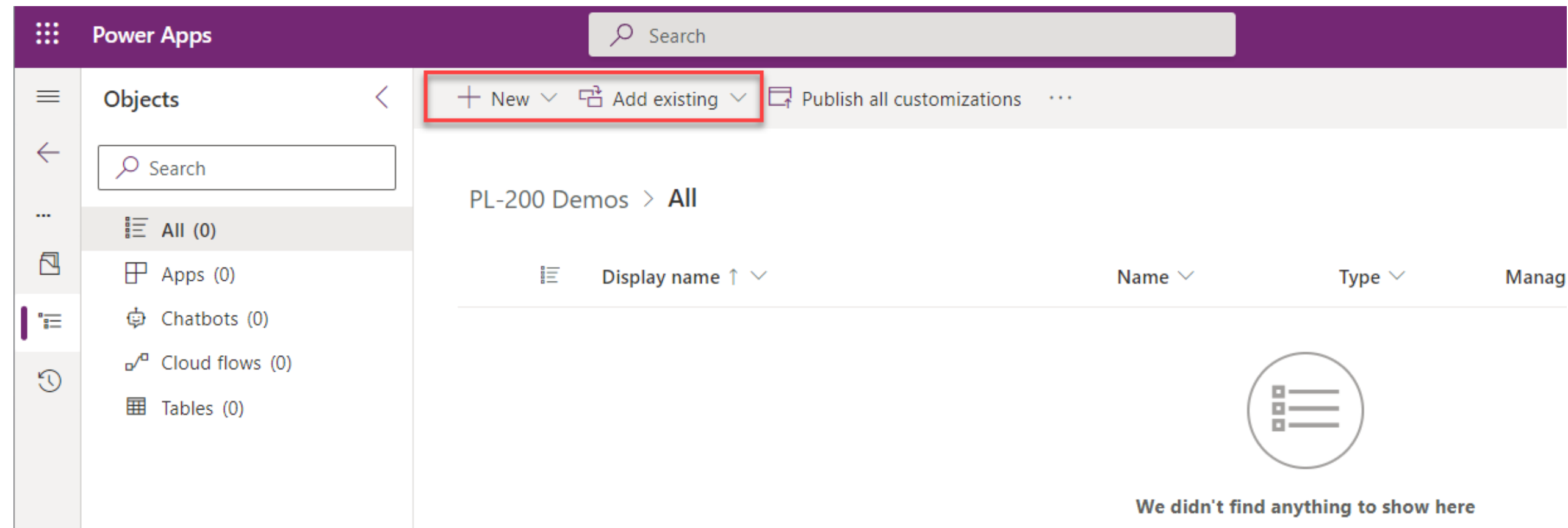
More options ▾



What's inside there?



Solution components



- Create new components in solutions
- Add existing components to solutions
 - Avoid **Include all objects** for Common Data Model tables (Account, Contact etc.)



Caring is sharing 🤝

Tips and tricks when working with solutions

- Solution aware versus non-solution aware components
- Always remember to KISS 🐱
 - Less is more
 - Dependencies and deployment order
 - Multiple managed layers
- Handle cloud flows and connection references via a dedicated solution and service account
 - Prevents flows from switching off post-deployment
- Consider automation for more complex scenarios:
 - [Power Platform Pipelines](#)
 - [Power Platform \(PAC\) CLI](#)
 - [Azure DevOps](#)



Lab 0 – CONFIGURE LAB ENVIRONMENT



In this lab, you will do the following:

- Redeem your Developer plan license for Power Apps, to create an environment to build your solution - this will be your Development environment.
- Set up an environment to deploy your solution - this will be your Production environment.
- Create a publisher and solution to store your components.
- Prepare your OneDrive for Business folder and upload the required document template.

Expected completion time: 15 minutes



Module 2: Model-Driven Apps



Model-Driven App Development

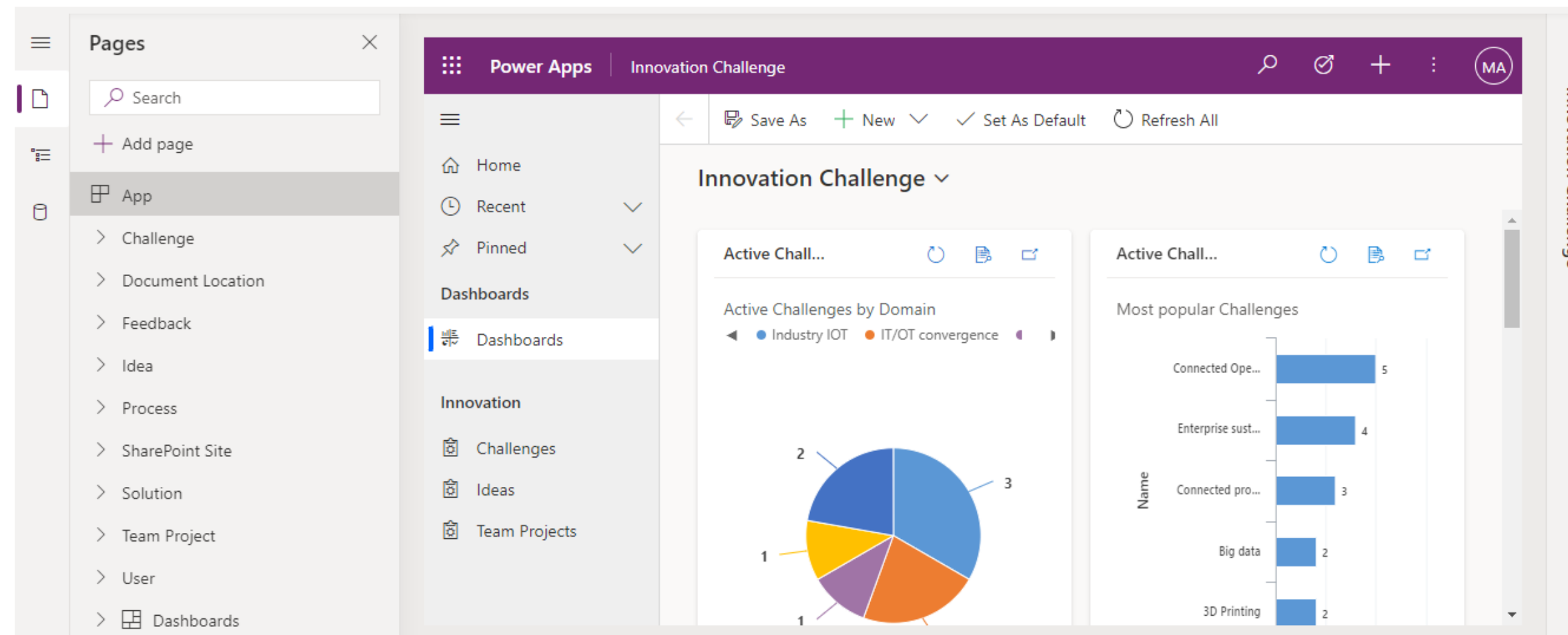
Before creating your app for the first time, consider the following:

- Working with the modern designer (and why the “classic” designer might be necessary)
- How the sitemap works
- Key app components:
 - **Tables**
 - **Forms**
 - **Views**
 - Charts
 - Dashboards



We like it modern

Modern app designer



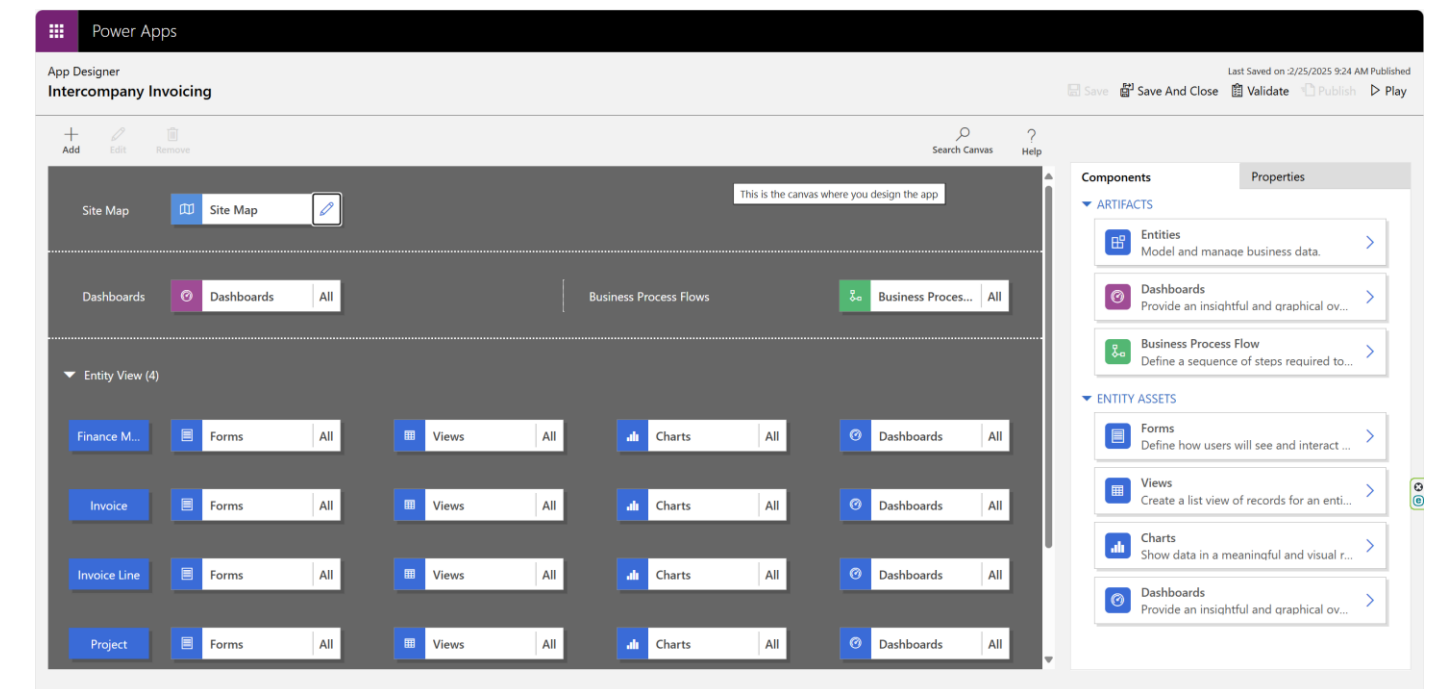
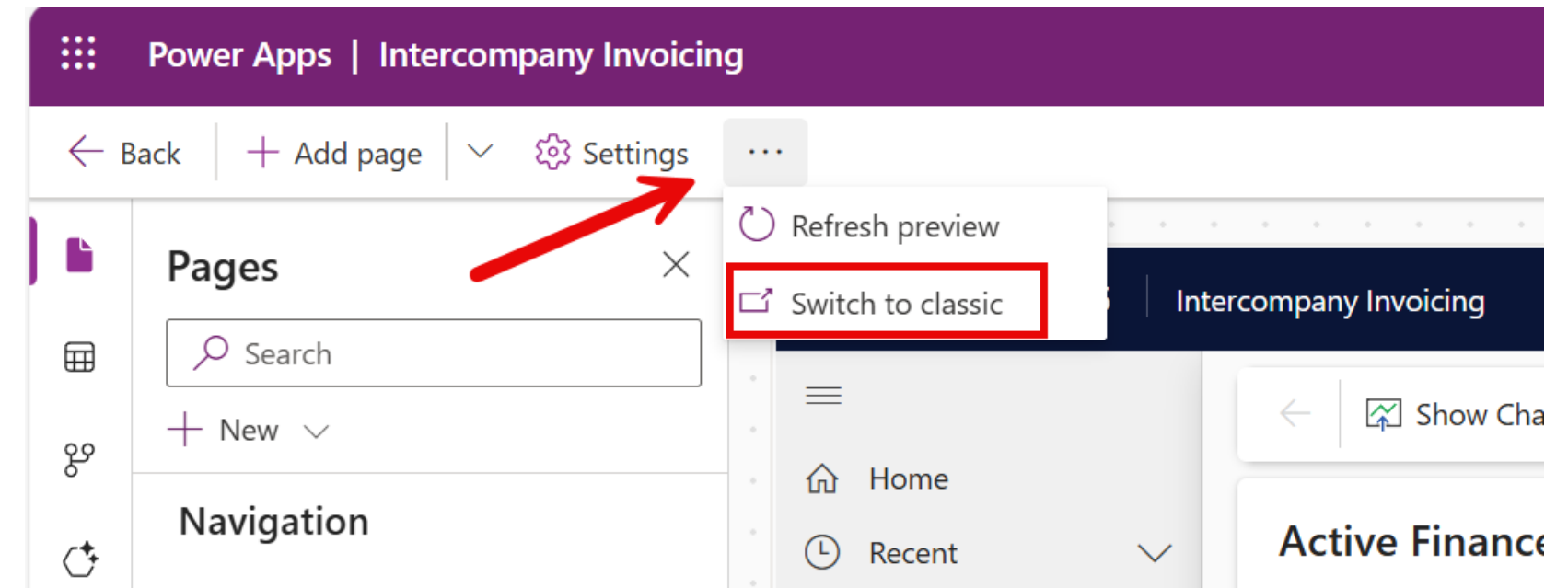
After modelling data, you build your app by selecting and setting up the pages you need in the App Designer.



Classics never die

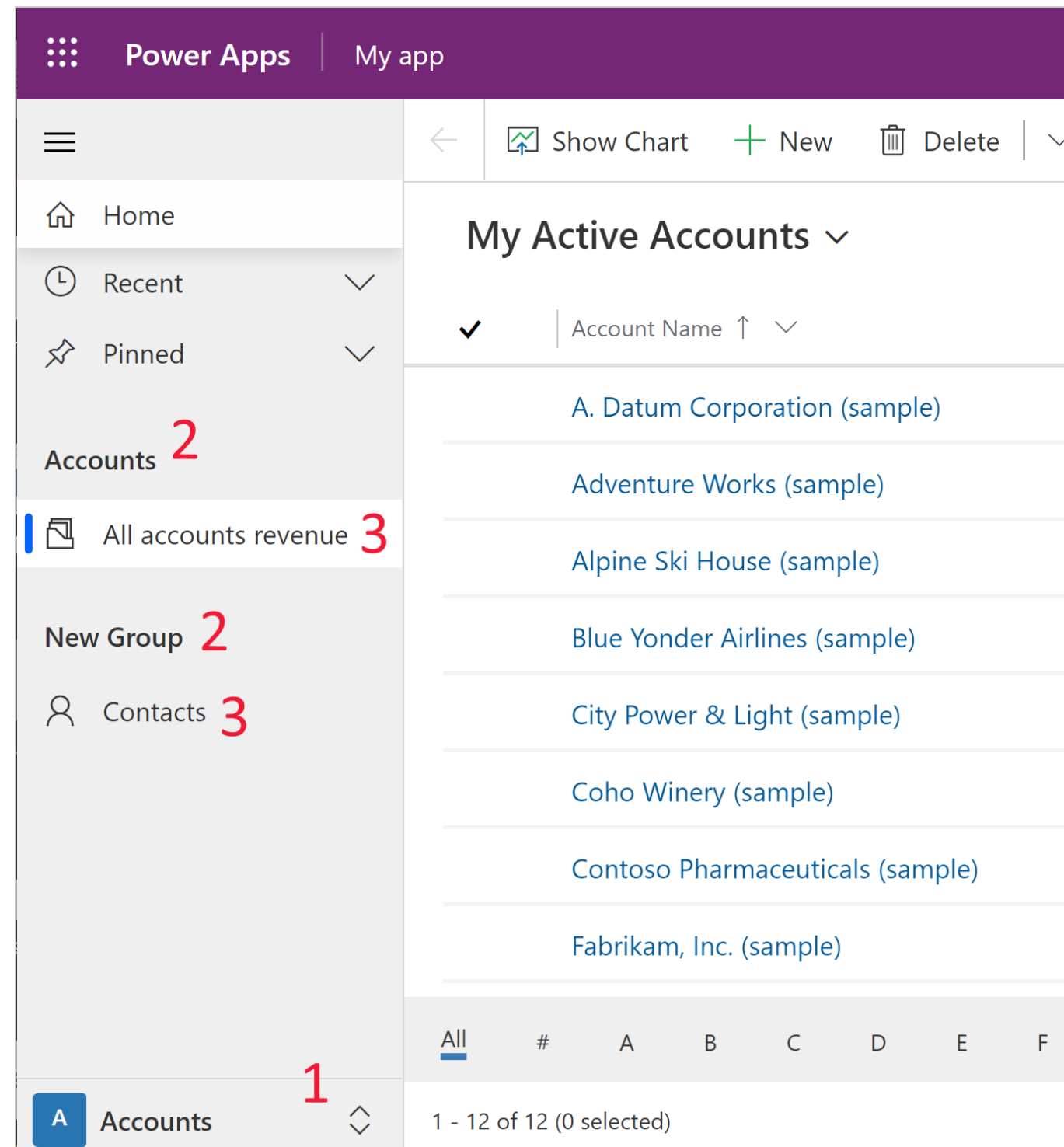
“Classic” designer

- The old way of building apps 🧓
- Still available for backwards compatibility; will eventually be removed.
- Can't do something in the modern designer? Go “classic”!
- Only use if strictly required.



Need some directions

Sitemap

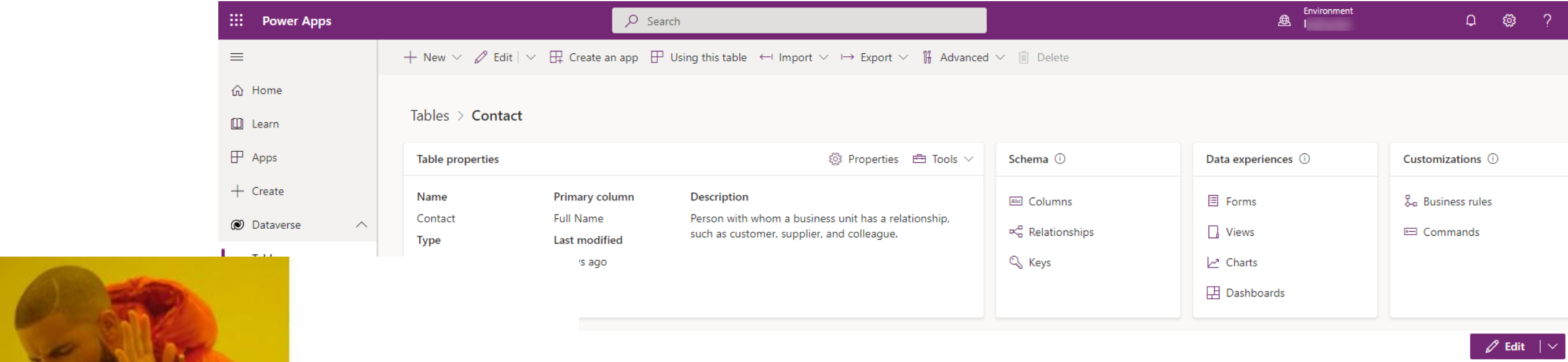


- Every model-driven app has a Sitemap that controls the left-hand navigation
- Structure of a site map:
 1. Area
 2. Group
 3. Subarea
- Pages are added as subareas



Relational data FTW

A table is a logical structure containing rows and columns that represents a set of data.



Tables



Entities

ie ▾	Email ▾	Full Name ↑ ▾	Business Phone ▾	+216 more ▾ +
imple)	someone_j@example.com	Jim Glynn (sample)	555-0109	
ample)	someone_d@example.com	Maria Campbell (sample)	555-0103	
is (sample)	someone_c@example.com	Nancy Anderson (sample)	555-0102	
e (sample)	someone_k@example.com	Patrick Sands (sample)	555-0110	
e (sample)	someone_h@example.com	Paul Cannon (sample)	555-0107	
ration (sample)	someone_i@example.com	Rene Valdes (sample)	555-0108	
aceuticals (sample)	someone_g@example.com	Robert Lyon (sample)	555-0106	
jht (sample)	someone_f@example.com	Scott Konersmann (sample)	555-0105	



Bling up your data model

Table characteristics

- Microsoft Dataverse is designed to let you quickly and easily create a data model for your application, based on the tables and the table metadata that you include in your app.
 - Display name
 - Plural name
 - Schema name
 - Primary column
 - Table type
 - Record ownership
 - Advanced options

New table

Use tables to hold and organize your data. Previously called entities

[Learn more](#)

Properties

Primary column

Display name *

Plural name *

Description

☐ Enable attachments (including notes and files) ¹

Advanced options ^

Schema name *

Type *

Record ownership *



What's in a name?

Primary column

Properties **Primary column**

Display name *

Name

Description

Advanced options ^

Schema name *

dem_ Name

Column requirement *

Business required

Maximum character count *

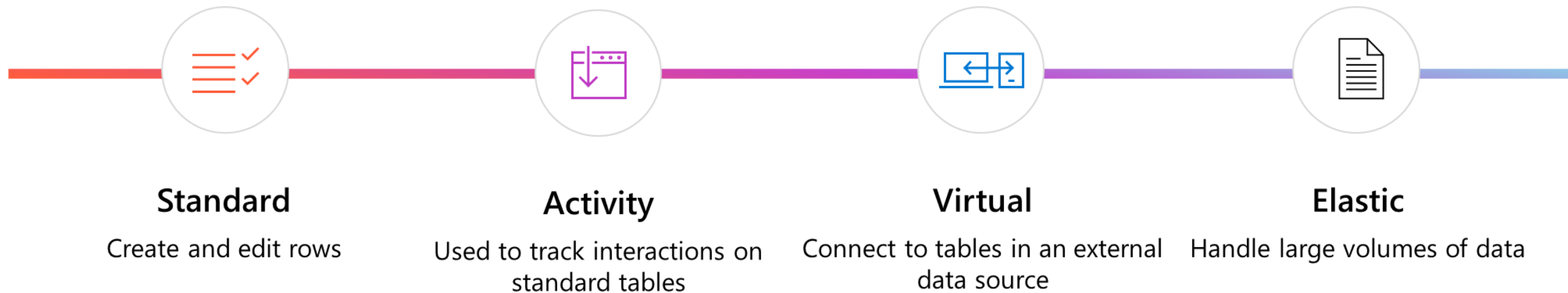
100

- Every table has a Primary Name column
- Single line of text
- Used by lookup columns to display the name of the related row
- Displayed in form heading and window title in model-driven apps



This is getting out of hand. Now there are four of them

Types of table



DEMO: Creating a custom table



Decisions, decisions...

Standard versus custom tables

- **Dataverse comes with several of standard tables that support core business application capabilities.**
 - To change the display name, you can edit the table. You don't have to create a new table.
 - You can't delete standard tables, but you can hide them. To hide a standard table, change the security role privileges for your users to remove the Read privilege for that table. This will remove the table from most parts of the application.





A columnar cornucopia

Columns are a way to store a discrete piece of information within a row in a table.

Power Apps

Search

Environment

Search

Home

Learn

Apps

Create

Dataverse

Tables

Choices

Dataflows

Azure Synapse Link

Connections

Custom Connectors

Gateways

Flows

Chatbots

AI Builder

Solutions

New column

Search

Tables > Account > Columns

Display name	Name	Data type	Managed	Customizable	Required	Searchable
(Deprecated) Process Stage	Stageld	Unique identifier	Yes	No	No	No
(Deprecated) Traversed Path	TraversedPath	Single line of text	Yes	Yes	No	No
Account	AccountId	Unique identifier	Yes	Yes	Yes	Yes
Account Name	Name	Single line of text	Yes	Yes	Yes	Yes
Account Number	AccountNumber	Single line of text	Yes	Yes	No	Yes
Account Rating	AccountRatingCode	Choice	Yes	Yes	No	No
Address 1	Address1_Composite	Multiple lines of text	Yes	Yes	No	Yes
Address 1: Address Type	Address1_AddressTypeCode	Choice	Yes	Yes	No	Yes
Address 1: City	Address1_City	Single line of text	Yes	Yes	No	Yes
Address 1: Country/Region	Address1_Country	Single line of text	Yes	Yes	No	Yes
Address 1: County	Address1_County	Single line of text	Yes	Yes	No	Yes
Address 1: Fax	Address1_Fax	Single line of text	Yes	Yes	No	Yes
Address 1: Freight Terms	Address1_FreightTermsCode	Choice	Yes	Yes	No	Yes
Address 1: ID	Address1_AddressId	Unique identifier	Yes	No	No	No
Address 1: Latitude	Address1_Latitude	Float	Yes	Yes	No	Yes

1,024 column table, here I come...

Add a column to a table

You can add columns when you create a new custom table, or you can add columns to an existing table at any time.

- Display name
- Descriptions
- Data type
- Format
- Behavior
- Required
- Searchable
- Schema name
- Advanced options

New column

Previously called fields. [Learn more](#)

Display name *

Description ⓘ

Data type * ⓘ

Format *

Behavior ⓘ

Required ⓘ

☒ Searchable ⓘ

[Advanced options](#) ^

Schema name * ⓘ



Gotta store 'em all

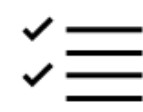
Column data types



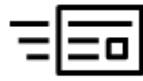
Single line of text



Currency



Choices



Multiple lines of text



Date and time



Yes/No



Whole number



Lookup



File



Floating point number



Customer



Image



Decimal number



Choice



You mean there's more??

Other column details



Column display formats



Column properties



Column requirement



Searchable



Column (field) security



Auditing



Description



DEMO: Working with custom columns



Perfectly formed

The Form Designer

The screenshot displays the Microsoft Dynamics 365 Form Designer interface. The central canvas shows a 'New Building' form with a 'Details' section containing fields for 'Building Name', 'Country', and 'Date Built'. Below this is a 'Value' section with 'Currency' (set to 'British Pound') and 'Price per Square Meter'. The 'Space' section includes 'Number of Floors', 'Last updated', 'Available Floor Space', and 'Last updated'. A 'Timeline' section is also present. To the right, a 'Related' section shows a table of buildings. The left sidebar shows a 'Tree view' with a search bar and a list of components: Information, Header, General, Details, Building Name, Country, Date Built, Value, Space, Timeline, Related, Floor, and Footer. The 'Building Name' component is selected. The right sidebar shows the 'Properties' tab for the 'Building Name' field, with options for 'Display options' (Table column, Edit table column), 'Label' (Building Name), and 'Formatting' (Form field width set to 1 column).

Model-driven forms are how data for a row is presented to the user during editing and viewing.



What's your type?

Form types

Main

Quick
View

Quick
Create

Card



You just need structure

Form structure

Model-driven app forms have a defined structure

1. **Header** – contains up to 4 columns.
2. **Body** – contains main content of the form.
3. **Footer** – no longer used.

The screenshot displays a model-driven app form for a fundraiser titled "Technical Education for Kids - Saved". The form is divided into a **Header** section and a **Body** section, both highlighted with red boxes.

Header Section:

- Left side: "Technical Education for Kids - Saved Fundraiser" and tabs for "General", "Donations", "Timeline", and "Related".
- Right side: A table with three columns: "Youth Category", "Fundraiser Goal", and "Total Donations". The values are "Youth", "£1,200.00", and "£425.00" respectively. A dropdown arrow is next to the "Total Donations" value.

Body Section:

- Name:** "Technical Education for Kids" (marked with an asterisk).
- Project name:** "Technical Education for Kids" with an "Edit" button.
- Owner:** "Joe Griffin (Available)" (marked with an asterisk).
- Story:** A text area containing the following text: "We're committed to ensuring all youth can learn the skills gained through computer science education including creativity, critical thinking, and problem-solving. A computer science education teaches real-world skills most in demand by employers and prepares young people for a wide variety of fields. Greater economic opportunity exists when young people have the digital skills needed to participate in a world transformed by technology. From digital literacy to computer science education, learning the digital skills critical to being future ready are often out of reach for the young people who need them most."
- Total Donations:** A summary card showing a green progress bar at 425, with the text "Last updated: 7/12/2025 8:11 AM".
- Video Link:** A card showing a video player with the URL "https://youtu.be/CFqTi-VMdoE" and a thumbnail image of a person at a computer.



You just need structure

Form structure

The body in a model-driven app form has a defined structure:

- 1. Tabs** – tabs can be 1, 2, or 3 column layout and contain sections.
- 2. Sections** – sections should be 1 column and contain controls.
- 3. Controls** – Controls are table columns or other specialized components.

Technical Education for Kids - Saved

Fundraiser

Youth Category

£1,200.00 Fundraiser Goal

£425.00 Total Donations

General Donations Timeline Related

Tabs

Form fill

Name

Technical Education for Kids

Project name Technical Education for Kids

Edit

Owner

Joe Griffin (Available)

Story

We're committed to ensuring all youth can learn the skills gained through computer science education including creativity, critical thinking, and problem-solving. A computer science education teaches real-world skills most in demand by employers and prepares young people for a wide variety of fields. Greater economic opportunity exists when young people have the digital skills needed to participate in a world transformed by technology. From digital literacy to computer science education, learning the digital skills critical to being future ready are often out of reach for the young people who need them most.

Total Donations

425

Last updated: 7/12/2025 8:11 AM

Control

Video Link

https://youtu.be/CFqTi-VMdoE

Boys & Girls Clubs of Am...

Section



I'm special. So special.

Specialized form components

Timeline

Properties

Display options

Name * ⓘ

Timeline

Record types to show ⓘ

☒ Activities

☒ Notes

Profile picture ⓘ

Default

^ Advanced

Quick entry record type ⓘ

Notes

Sort default

↓ Descending

Display layout ⓘ

Roomy

Records shown on page ⓘ

10

Filter pane

☒ Enable filter pane

☐ Expand filter pane by default ⓘ

Edit filter pane

Additional settings

☐ Hide Timeline label

☒ Hide the date portion for any datetime in the past 24 hours

☒ Enable search bar ⓘ

☐ Expand all records by default

☐ Enable "What you've missed" summary ⓘ

Move timeline to a new tab for mobile apps ⓘ

Default

^ Activities

Activity types

Appointment ✓

Email ✓

Fax

Letter

Phone Call ✓

Recurring Appointment ✓

Social Activity ✓

Task ✓

Teams chat ✓

Sort activities by ⓘ

Last Updated

Create activities using ⓘ

Quick create form

Open activities using ⓘ

Main form

Activity rollup type ⓘ

Extended

- Timeline
- Subgrid
- Quick view
- Reference panel
- Timer control
- IFrame

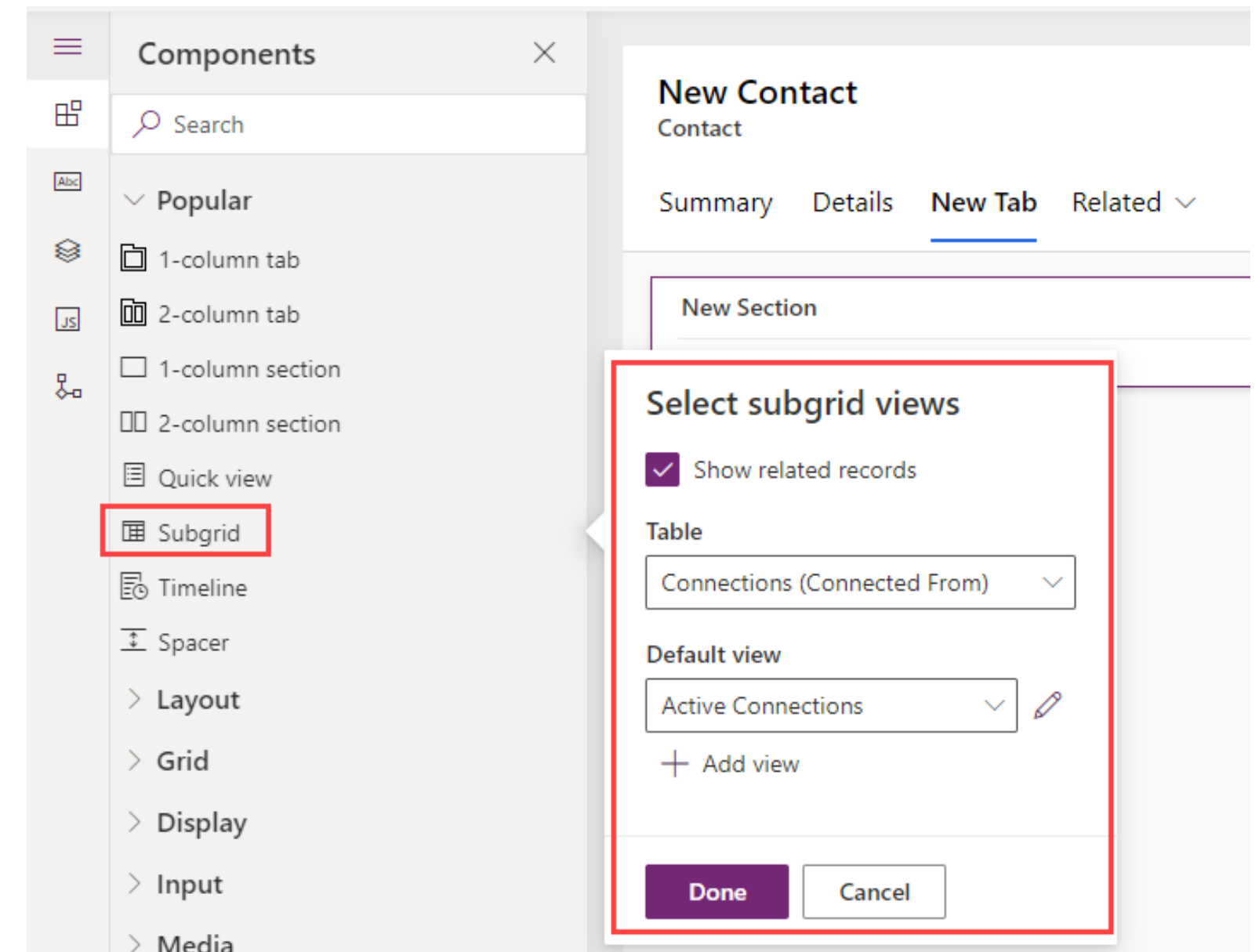


We are family

Show related data in a form

Subgrid Control

- Shows one-to-many and many-to-many relationship rows.
- Select the related table.
- Select a view to use to display data.



We are family

Show related data in a form

The screenshot shows a 'New Contact' form. On the left, a 'Components' sidebar lists various UI elements: Layout, 1-column section, Spacer, Grid, and Subgrid. The main form area is titled 'New Contact' and contains four input fields: First Name, Last Name, Email, and Company. Below these fields is a section titled 'Addresses' which is currently empty, showing a 'No data available' message. A '+ New Address' button is visible in the top right of the Addresses section.

Quick view form control

- Shows columns from a many-to-one relationship.
- Select the lookup column.
- Select the Quick View form for the related table.
- Columns, 1-column sections, and subgrid controls can be added to Quick View forms.



DEMO: Working with the form designer



Quickly now

Quick Create form

Configure Quick Create

- Enable Quick Create for a table in table properties.
- Create a new form.
- Quick Create forms have a single 1-column tab with 3 sections.
- Can only add columns to a Quick Create form.

For this table

- ☒ Apply duplicate detection rules ⓘ
- ☒ Track changes ¹ ⓘ
- ☐ Provide custom help ⓘ
- ☐ Audit changes to its data ⓘ
- ☒ Leverage quick-create form if available ⓘ
- ☒ Enable long term retention ⓘ

Help URL

Quick Create: Contact

×

New Section

New Section

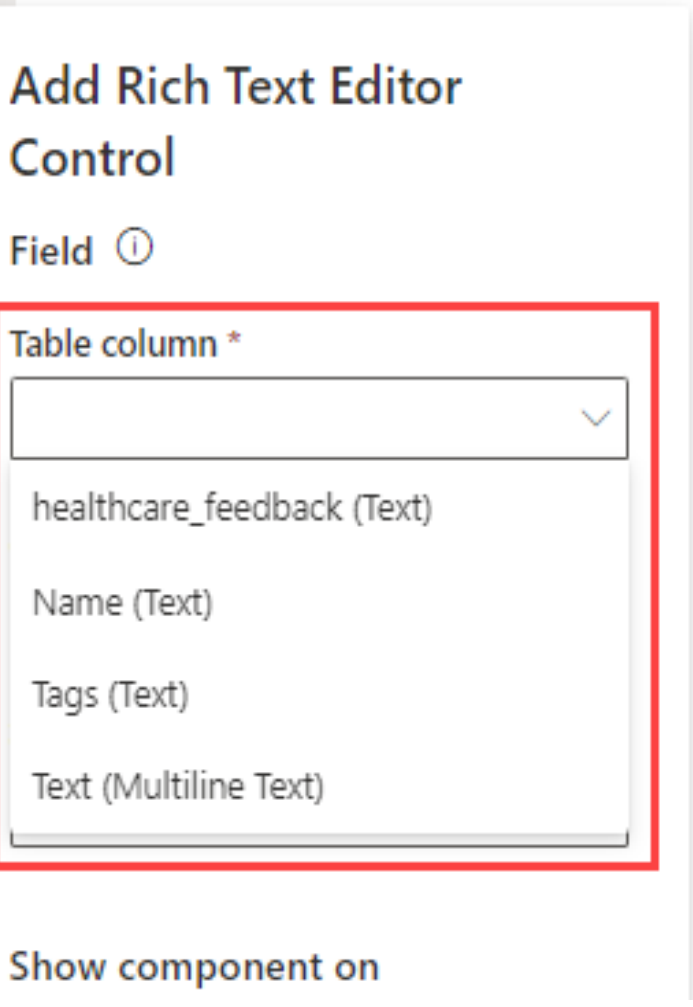
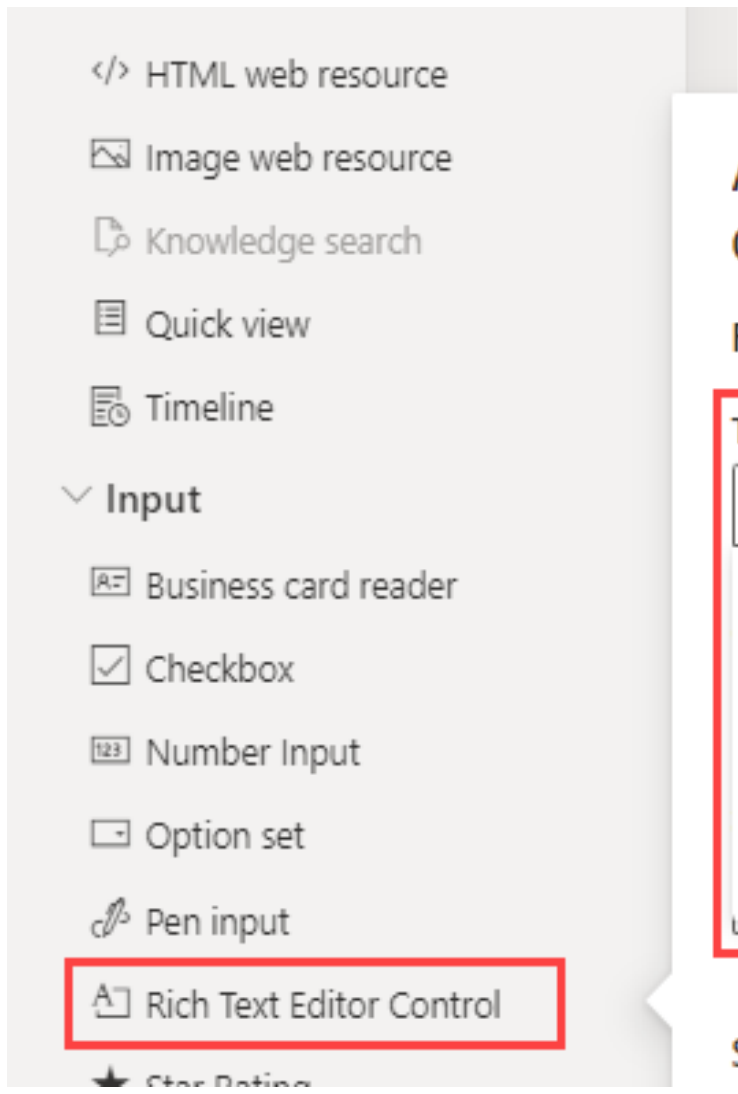
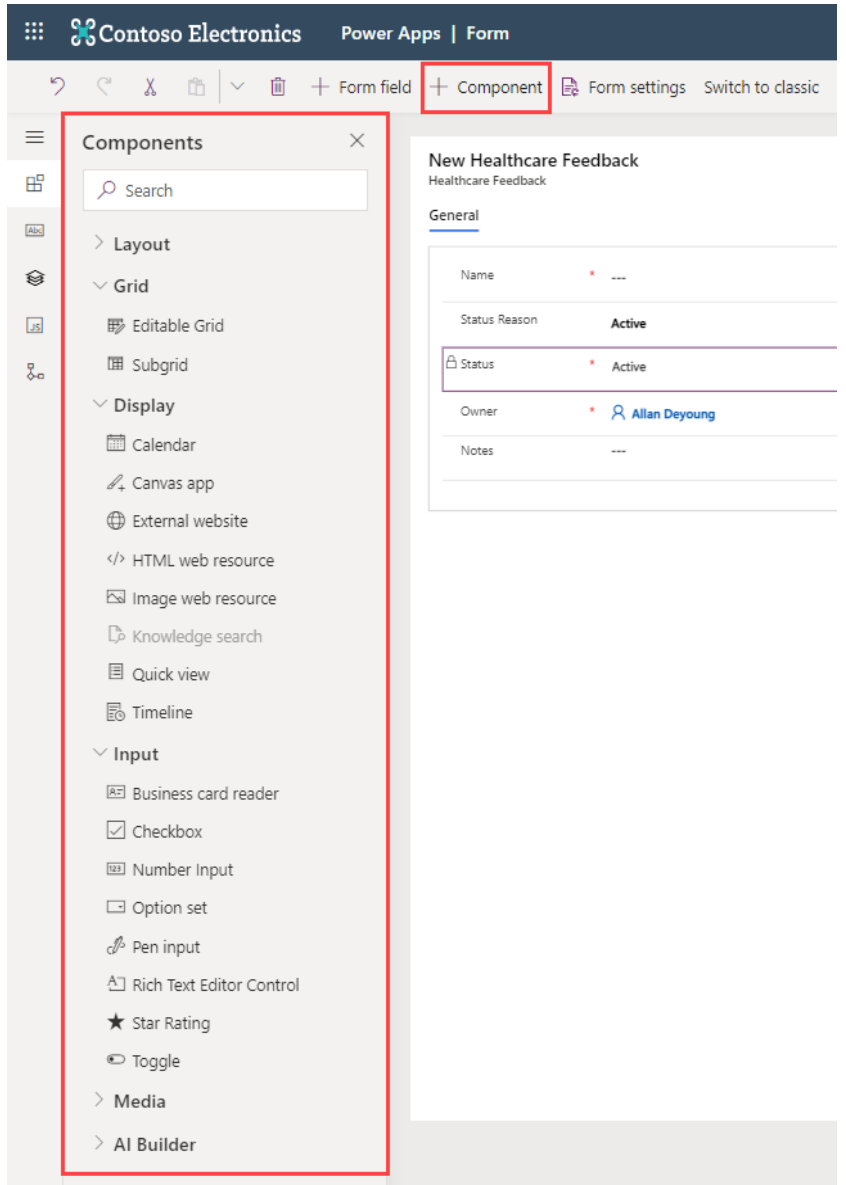
New Section



FCC FTW

Form component controls

Components can be added to columns on the forms to display how the control is displayed

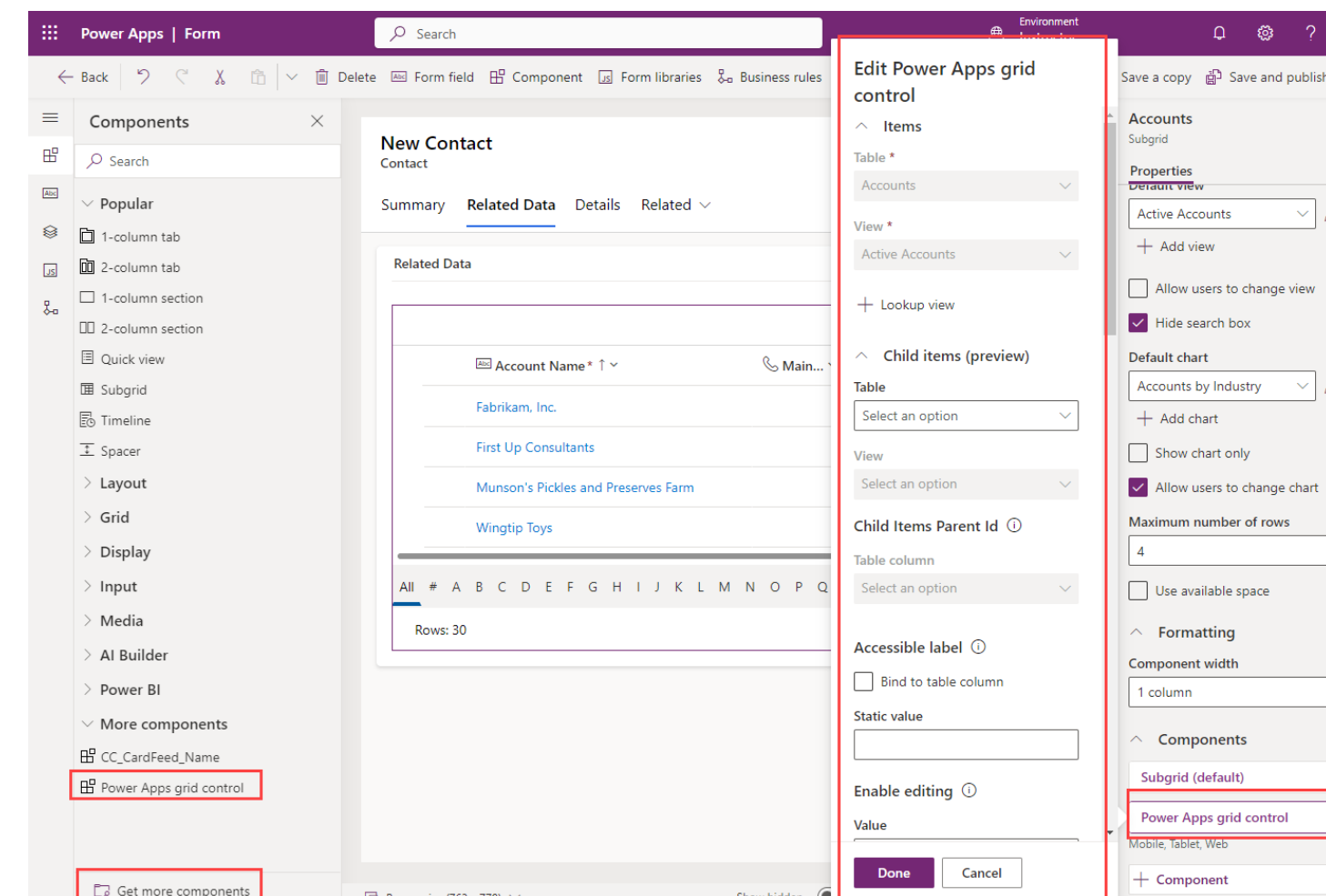


Hey, that's pretty grid

Power Apps grid control

By default, grids are read-only

- Editable grids and Power Apps grid controls are special controls for editing columns in a view without needing to open a form.
- Add the Power Apps grid control to the form.
- Add Power Apps grid control as a component to a subgrid.
- Features:
- Read-only or Editable.
- Control reflow behaviour.
- Filtering.
- Sorting.
- Jump bar.
- Data type icons.



Hey, that's pretty grid

Power Apps grid control

Security roles
Form order
Fallback forms
Form access checker

Form order for Account table

When multiple forms are enabled for a user, choose the order in which they will appear. Ideally, we recommend only having one form per security role. [Learn more](#)
Drag and drop to change the order or use the keyboard (Tab/Tab+Shift to navigate the list, Space to select/release a form, Up/Down to move the form).

Choose a form set

Main Form

Name

Account

Account for Interactive experience

Information

Save and publish
Cancel

- Main forms can be associated with security roles.
- Users will see the form that matches their security role.
- If the user has more than one form, the Form order defines which form the user will see.
- If, through security, the user has no forms then the user will see the form designated for fallback.



Let's get the form party started

Configure multiple forms

- Only Main forms can be assigned to security roles.
- When a user has access to multiple forms, a form selector will be available near the top of the form. If a user has access to only a single form for a given table, there will be no form selector visible.
- You can designate a main form as inactive. This will make it not visible to all users, regardless of security roles.
- Controlling user access to forms is not necessarily a secure means to prevent access to data. Sometimes users have other ways to interact with data such as advanced find or automation.



What a view!

Views Overview

- **Views are lists of data from a Dataverse table**
 - Columns from table
 - Columns from many-to-one relationships
 - Sorting
 - Filtering

Table columns

Building Related

Search

+ New table column

- Created By
- Created By (Delegate)
- Created On
- Currency
- Date Built
- Exchange Rate
- Modified By
- Modified By (Delegate)
- Modified On
- Owning Business Unit
- Price per Square Metre
- Price per Square Metre (Base)

Building Name ↑	Country	Number of...	Available F...	Owner
Building A	Sweden	3	300	
Building B	Denmark	1	25	
Building C	Sweden	1	200	

My Buildings

View

Name *

My Buildings

Description

Sort by ...

↑ Building Name

Then sort by ...

Filter by ...

Status is 'Active'

Owner is current user



What a view!

Public vs Personal views

Tables > Building > Views ▾

Name ↑ ▾	View type ▾
Active Buildings	Public View
Building Advanced Find View	Advanced Find View default
Building Associated View	Associated View default
Building Lookup View	Lookup View default
Buildings with floors	Public View
Inactive Buildings	Public View
My Buildings	Public View
Quick Find Active Buildings	Quick Find View

Building Details ▾

My Views

Building Details

System Views

My Buildings Default

Active Buildings

Inactive Buildings

 Set current view as my default

- **Public (System) views are managed in the Maker portal**
 - Public views are not controlled by security
 - Public views can be turned off (deactivated)
 - Public views can be included/excluded in an app
- **Personal (My) views are created in a model-driven app**
 - Personal views can be shared with users and teams



Chancellor, Views are our specialty

Specialty views

The screenshot displays the Microsoft Dynamics 365 user interface. At the top, the header includes the Microsoft Dynamics 365 logo, the user name 'MOD Administrator' for 'Contoso', and a help icon. Below the header is a ribbon with tabs: 'FILE', 'ADVANCED FIND', and 'LIST TOOLS'. The 'LIST TOOLS' tab is active, showing a 'CONTACTS' sub-tab. The ribbon contains various action buttons grouped into 'Actions', 'Collaborate', and 'Process'. The main area shows a list of contacts with columns for 'Full Name' and 'Business Phone'. The list includes four entries: Adrian Dumitrascu, Adrienne McMillan, Alex Simmons, and Alex Wu. On the right side, there is a 'Related - Common' sidebar with a list of related entities: Activities, Social Profiles, Contacts, Connections, Entitlements, Actuals, Resource Preferences, Opportunity Lines, Quotes, and Quote Lines.

Full Name ↑	Business Phone
Adrian Dumitrascu	768-555-0156
Adrienne McMillan	
Alex Simmons	1-555-555-5554
Alex Wu	555-123-4879

There are four types of specialty view; Lookup, Associated, Quick Find, Advanced Find (legacy)



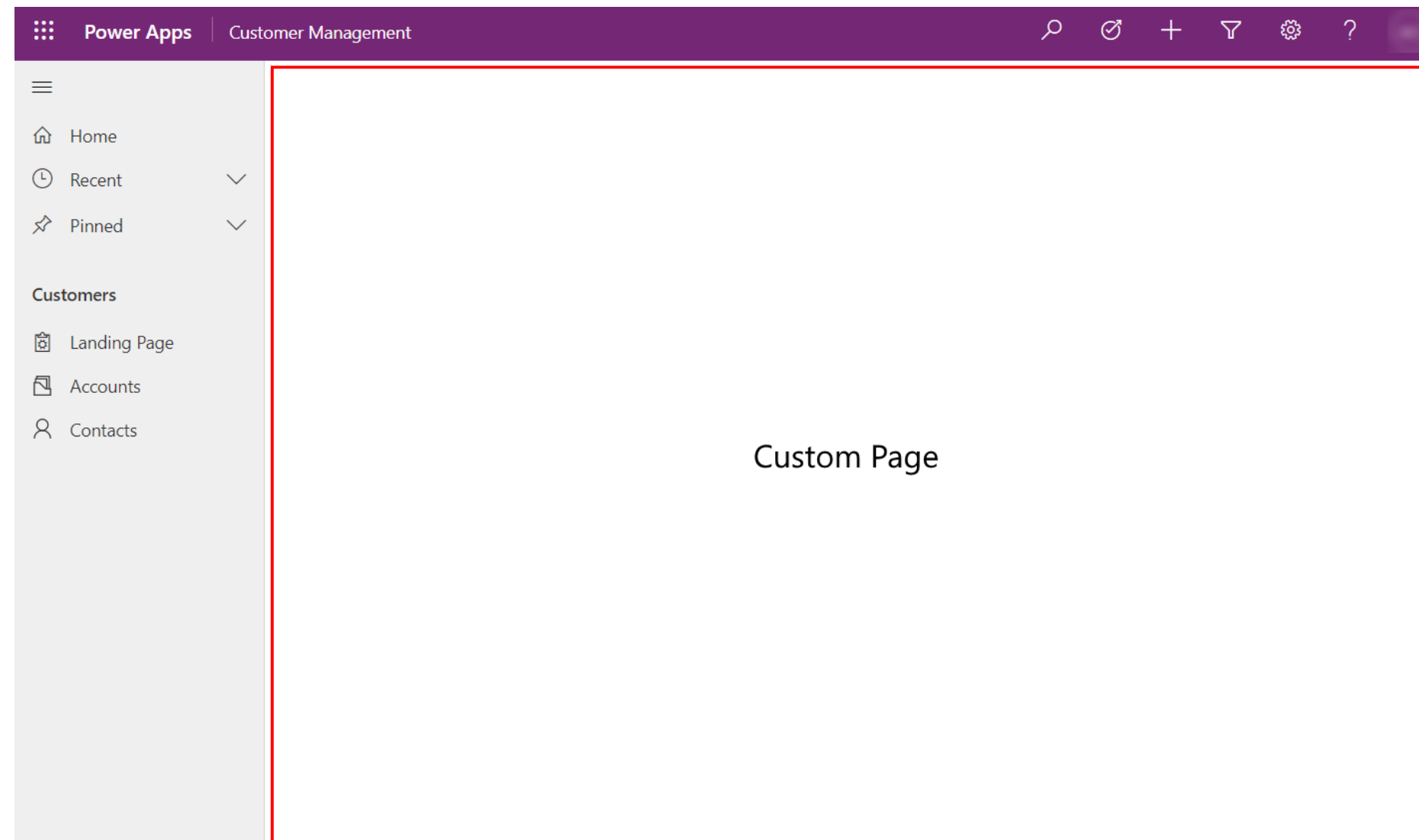
DEMO: Working with views



A beautiful, blank canvas



Custom pages



What is a custom page?

- A new page experience using the canvas app capabilities.
- Added to the site map in a model-driven app.
- Uses connectors to access data external to Dataverse inside a model-driven app.
- Recommend to use instead of embedding canvas apps.
- Separate component in a solution.

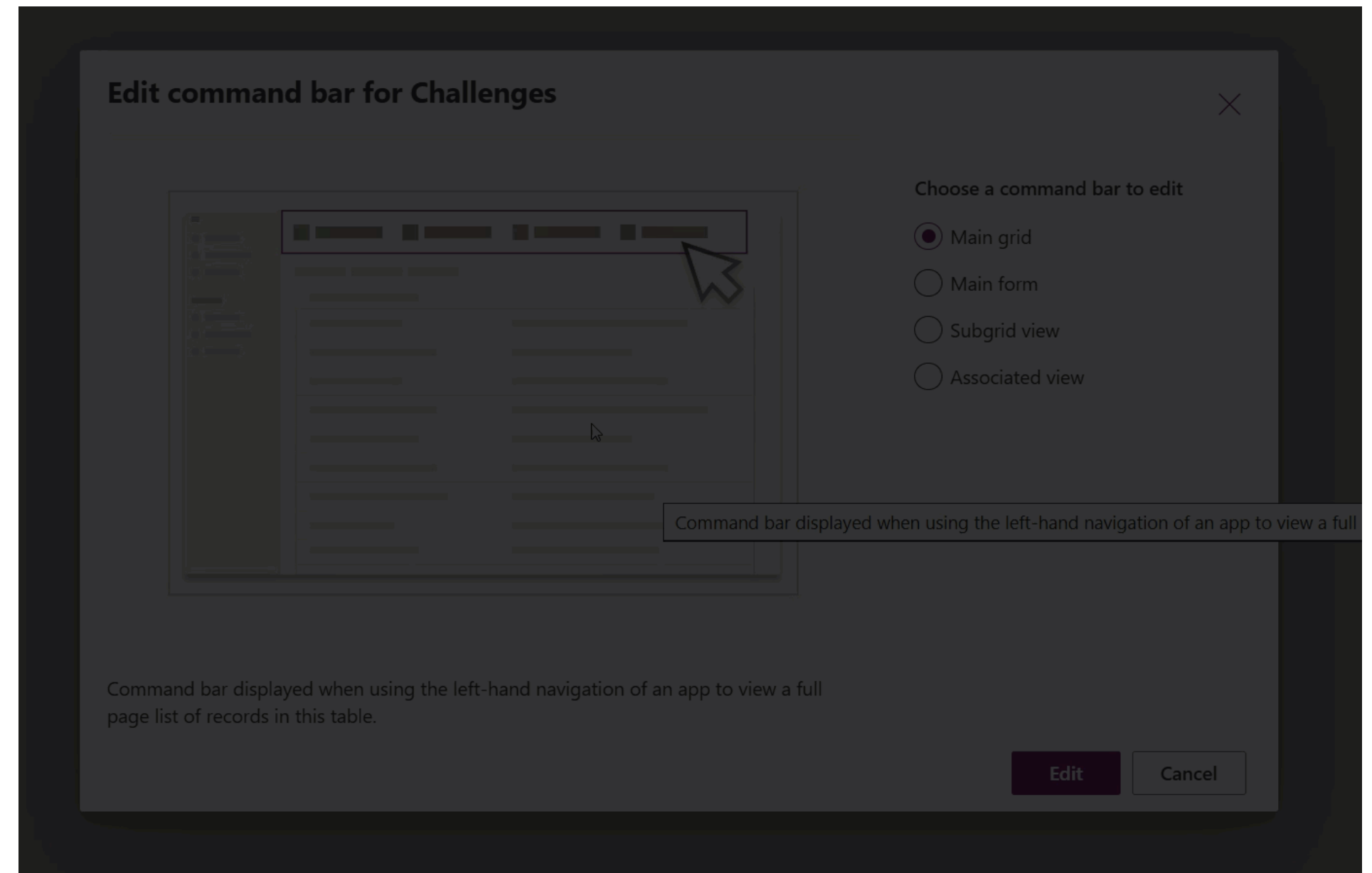


On my command, unleash Hell

Command bar

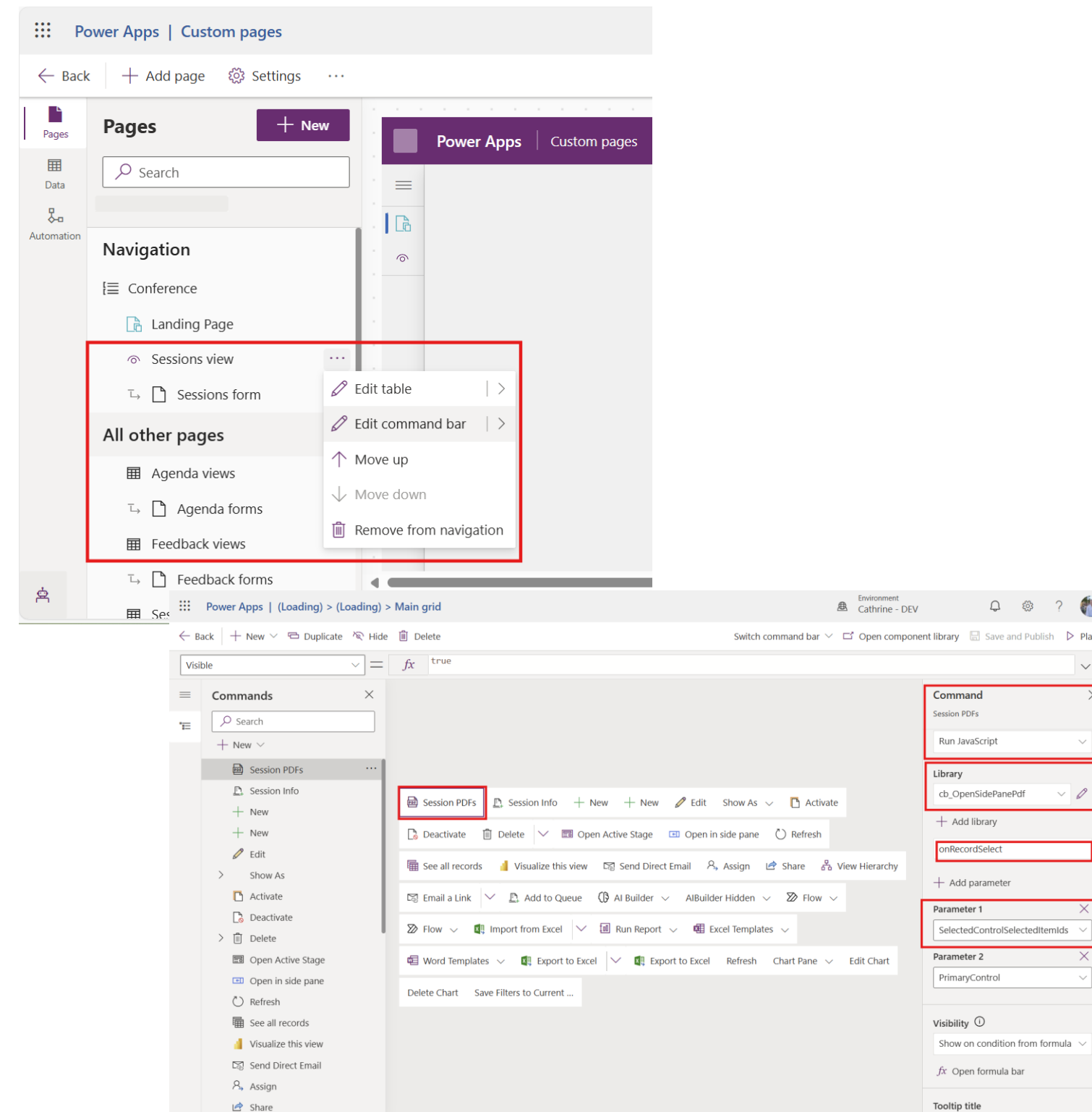
Locations

1. Main grid
2. Main form
3. Subgrid view
4. Associated view



What is this I don't even

Creating a Command



1. Select your table and the three dots on the right
2. Select «Edit Command bar»
3. Choose whether the button should be visible on the form or from the Main Grid
4. Select «+New» - «Command»
5. On the right side, provide a name for the button
6. Select «Run JavaScript»
7. Select or search for the web resource as the library
8. Set the function name «JavaScriptFunction»
9. Set parameters
10. Save and publish – Play the app to test



RIP Ribbon Workbench



Command bar designer

► **MODULE 2**

LAB 1

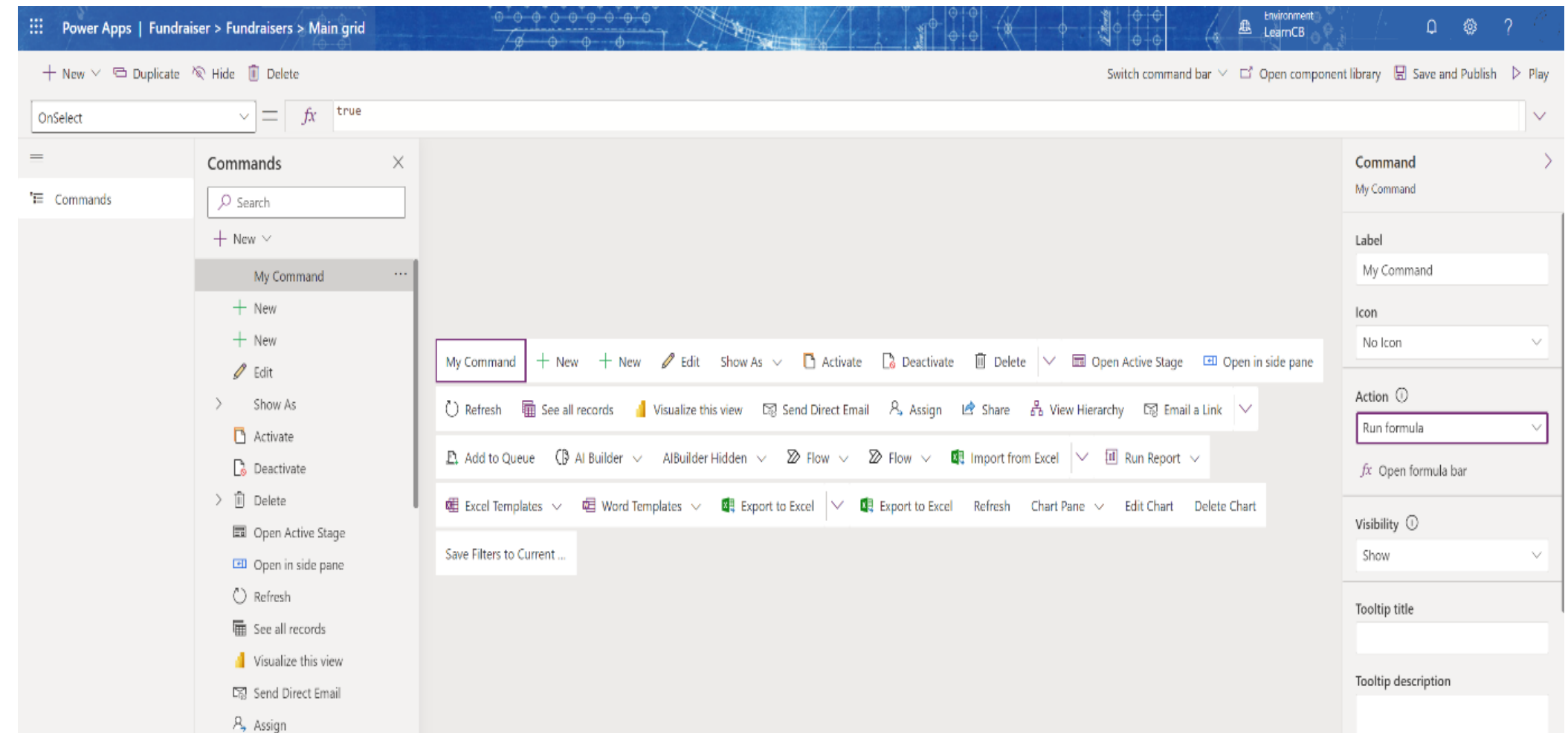
MODULE 3

LAB 2

LAB 3

Edit in model-driven app

1. Choose command bar
2. Visual representation
3. Add new commands
4. Modify existing commands



Select Edit
command
bar

Choose
command
bar to edit

Add a
button

Drag button
to desired
location

Configure
visual
display

Configure
action

Manage
visibility



Lights

Configuring actions

Camera

Action

Power Fx

- Properties
 - OnSelect
 - Visible
- Creates a component library
 - Ensure this is present in your solution
- Example use cases
 - Access related rows
 - Create and update rows with Patch

JavaScript

- Requires JavaScript web resources



DEMO: Working with the modern command designer



Before you get too excited...

Power Fx commands - Considerations and Limitations

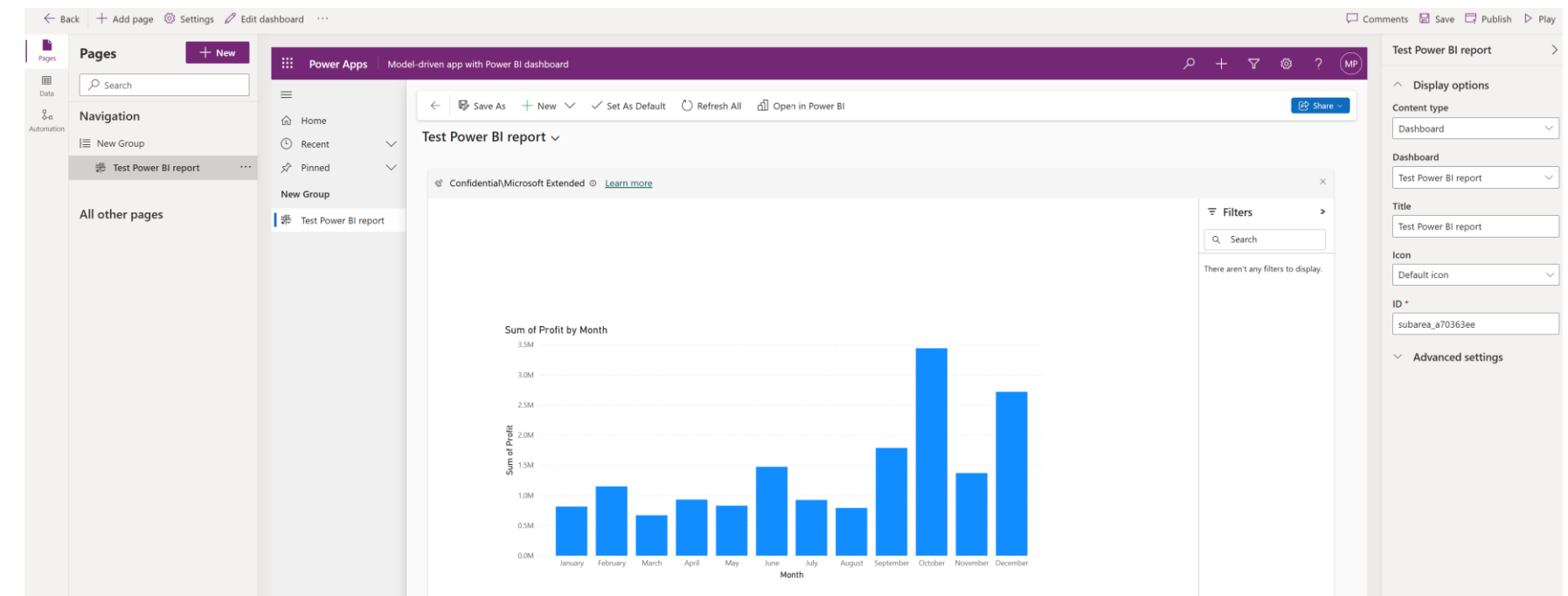
- Visible and OnSelect are the only supported properties
- Specific functions are not supported
- Unable to execute Power Automate cloud flows
 - Consider a Patch() operation and Dataverse connector trigger
- No support for additional data sources
- Power Fx commands do not work in the Dynamics 365 App for Outlook
 - Consider using JavaScript in this scenario



Make that data look snazzy

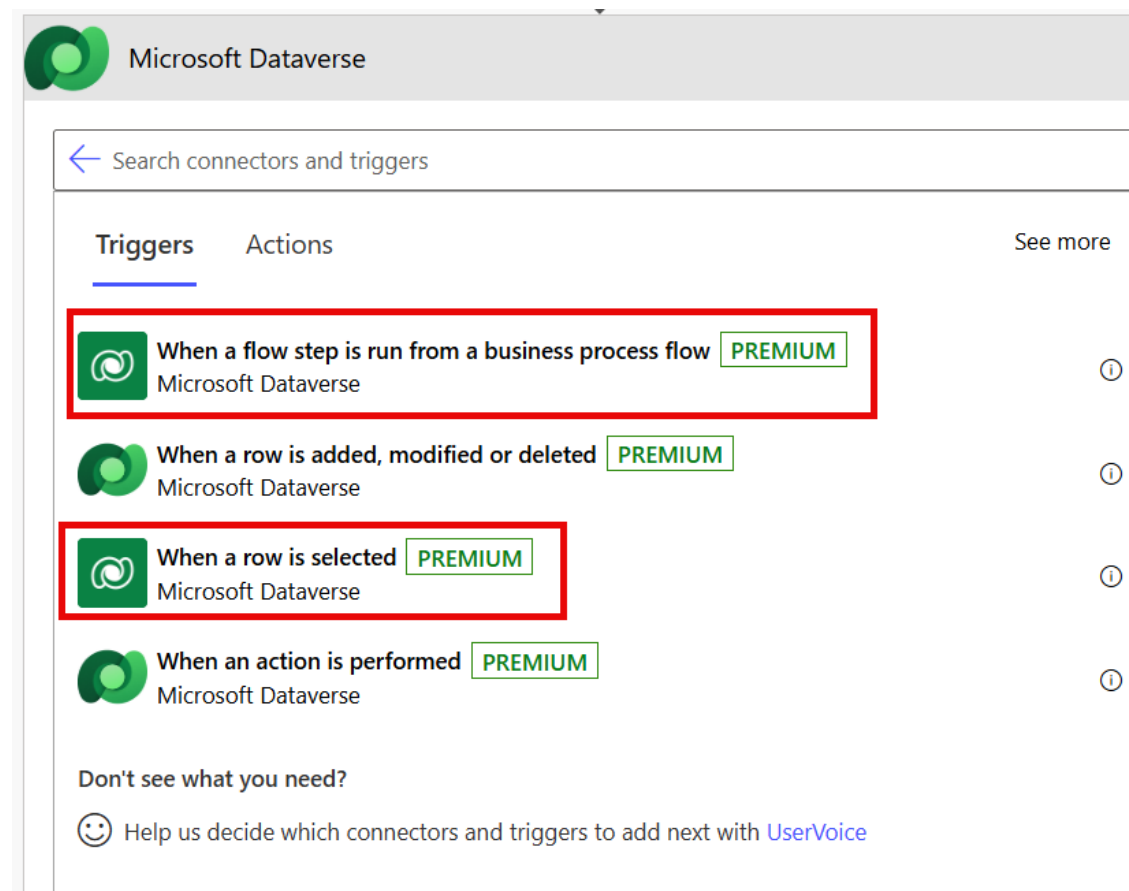
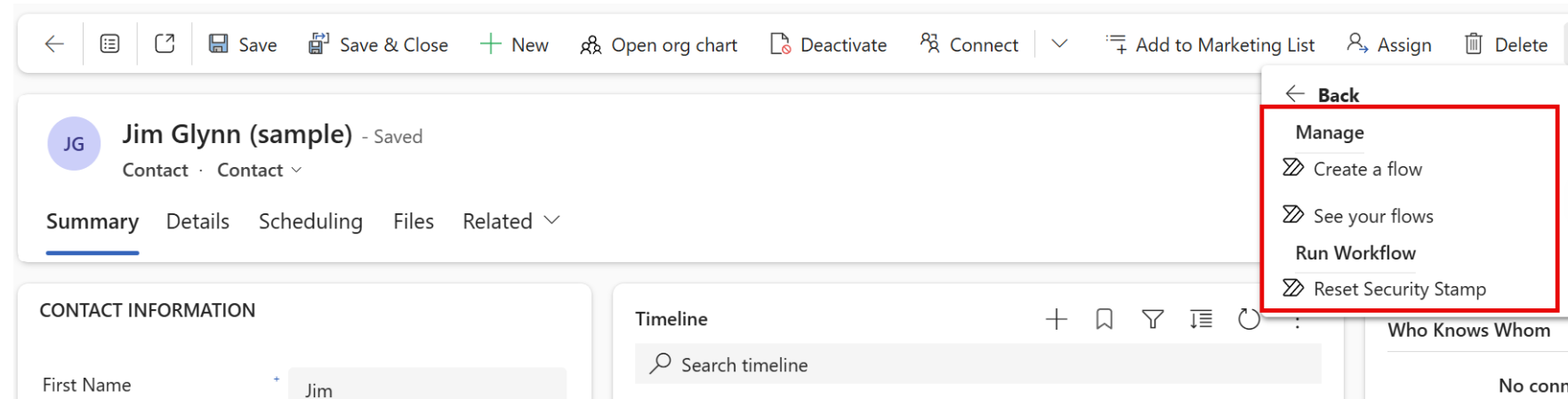
Additional Integration Options – Power BI

- Embed Power BI content in your apps.
 - Model-Driven
 - Report
 - Dashboard
 - Canvas Apps / Custom Pages
 - Tiles from dashboards only
- Consider ALM implications
 - Separate workspaces
 - Use environment variables



Go with the flow

Additional Integration Options – Power Automate



- Execute a cloud flow from
 - Any model-driven app row
 - Within a Business Process flow step
 - Canvas app / custom page:
 - *'My Flow'.Run(parameter1, parameter2)*
- 30+ actions targeting Microsoft Dataverse
- Ideal for more complex scenarios:
 - Complex iterations
 - Working with XML / JSON data
 - Child flows



I've got the Power (Fx)

Additional Integration Options – Power Fx

- Power Fx functions:
 - Solution-aware component
 - Support input / output parameters.
 - Can be bound to a record.
 - Some functions not supported
- Low-code plug-ins (preview)
 - Low-code mechanism to trigger synchronous logic in Dataverse
 - Work with via the Dataverse Accelerator app

New function ×

Use Power Fx functions to create reusable custom logic. Functions can be called from apps, flows, and other items. [Learn more](#)

Display name *

Validate Email function

Description *

Function to perform Regex validation for emails.

Parameters

Name *

StringIn

Data type *

String



+ New input parameter

Name *

regexMatch

Data type *

Boolean



+ New output result

Table references ⓘ

Select table references for the formula

Formula * ⓘ

```
{
  regexMatch : IsMatch(StringIn, "[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.\.[a-zA-Z]{2,}$")
}
```

Save

Cancel



Client-side scripting

JavaScript in model-driven apps via the Xrm object model

▶ **MODULE 2**

LAB 1

MODULE 3

LAB 2

LAB 3



What is the Xrm API?

A JavaScript object model in model-driven apps. Interact with forms, controls and records at runtime.



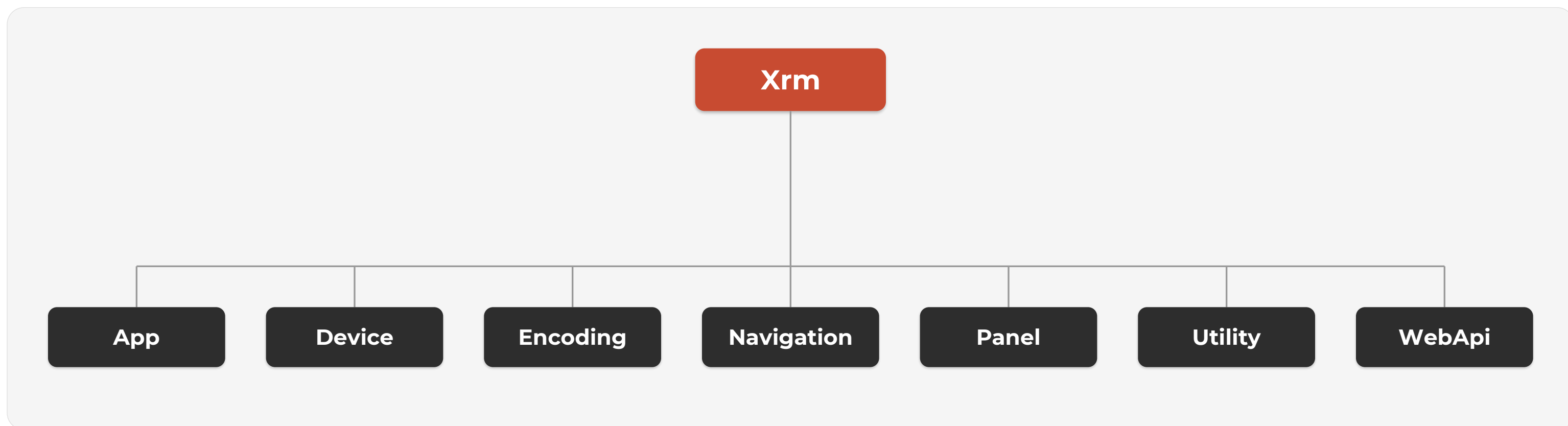
Key namespaces

formContext for form data & UI, Xrm.WebApi for CRUD, Xrm.Navigation for dialogs & URLs.



Why it matters

Validate fields, show/hide sections, auto-populate data and call services — before a record is saved.



AI Gore invented it (kind of)

Web resources

- Upload files to Dataverse
- Virtual file system
- Use in forms and apps
- Use as JavaScript libraries

Types:

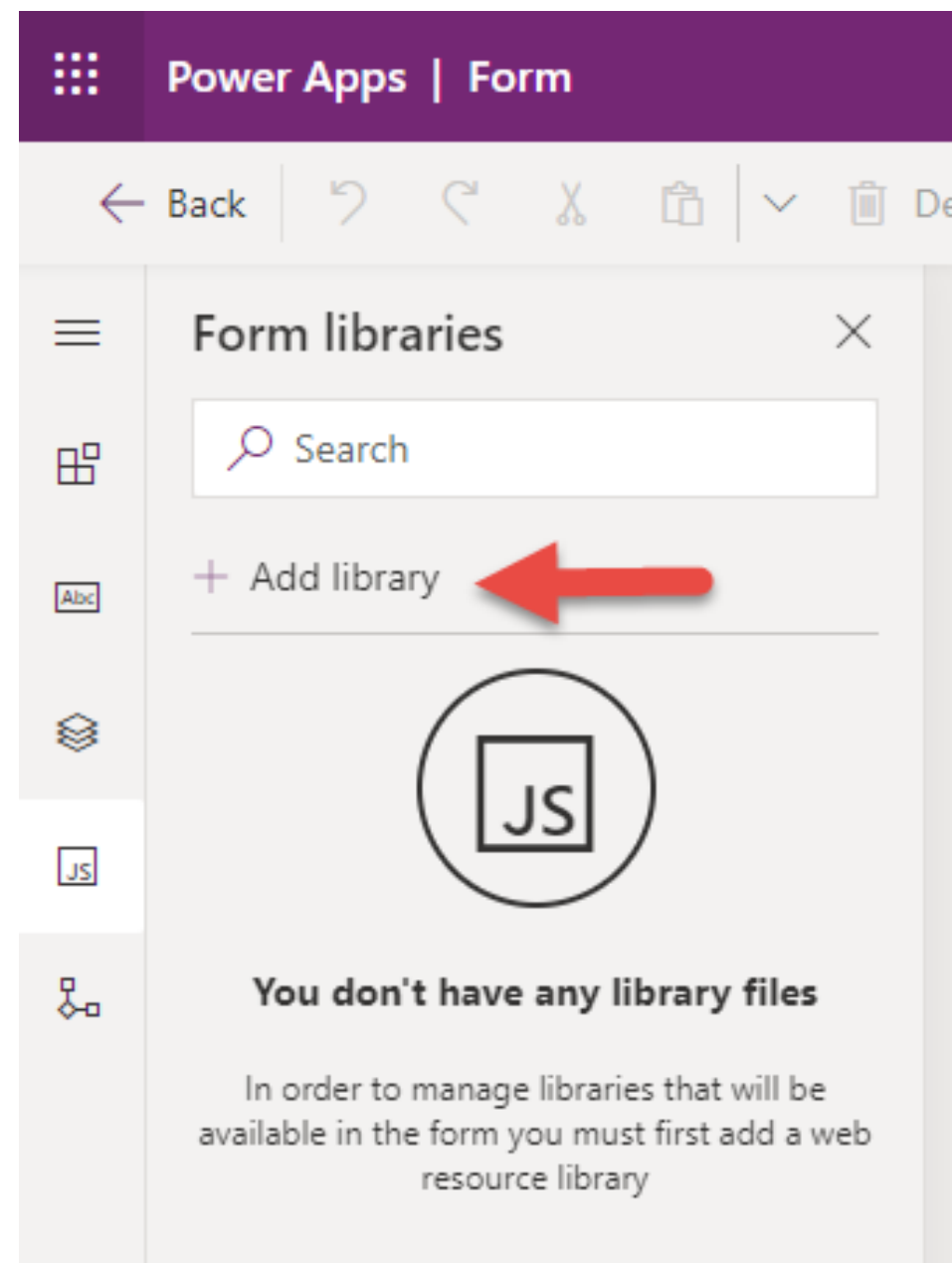
- Script (Jscript)
- Webpage (HTML)
- Style Sheet (CSS)
- Data (XML)
- **Images:**
 - PNG
 - JPG
 - GIF
 - ICO



Totes legal code injection

Upload scripts

To use client scripting on a form the script must first be uploaded as a JScript web resource



Upload a file *

contoso_common_libs.js

Display name

Name *

Type *

Description

Advanced options ^

Language

URL

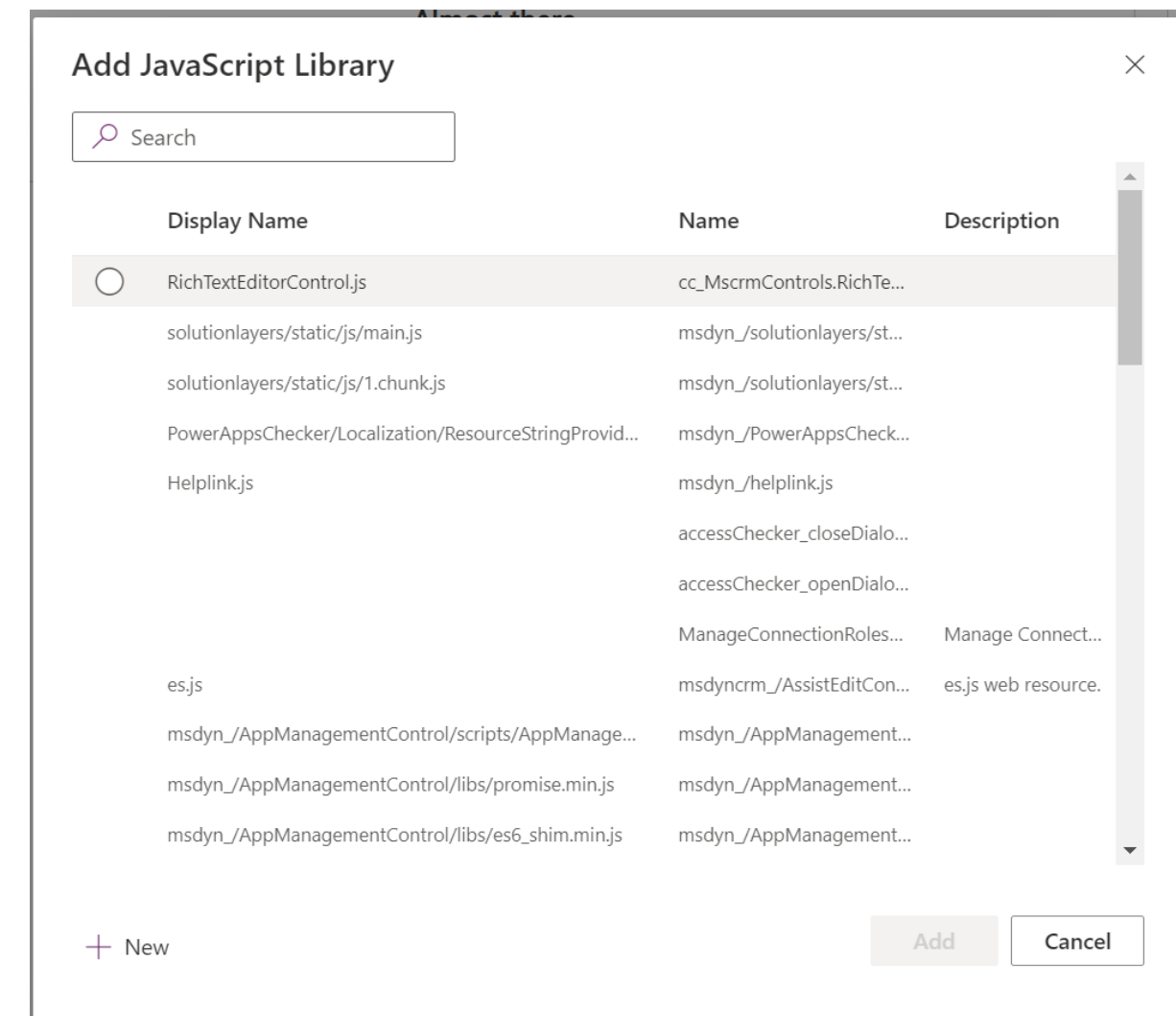
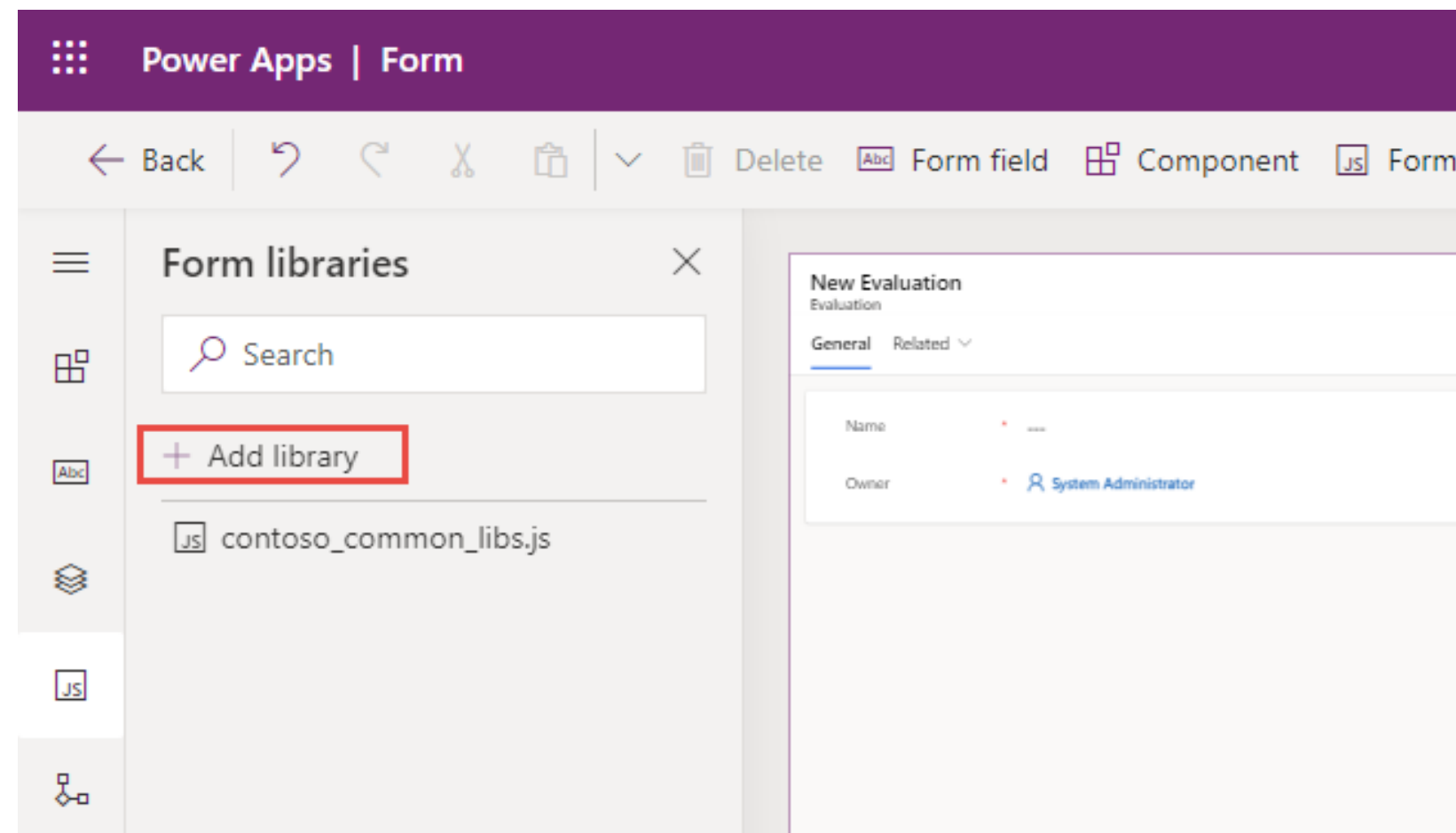
https://orgfbb147bc.crm.dynamics.com/WebResources/contoso_common_libs.js



A few itty, bitty steps

Using client script libraries

Once configured as a script web resource, client script libraries can be associated with commands and form events

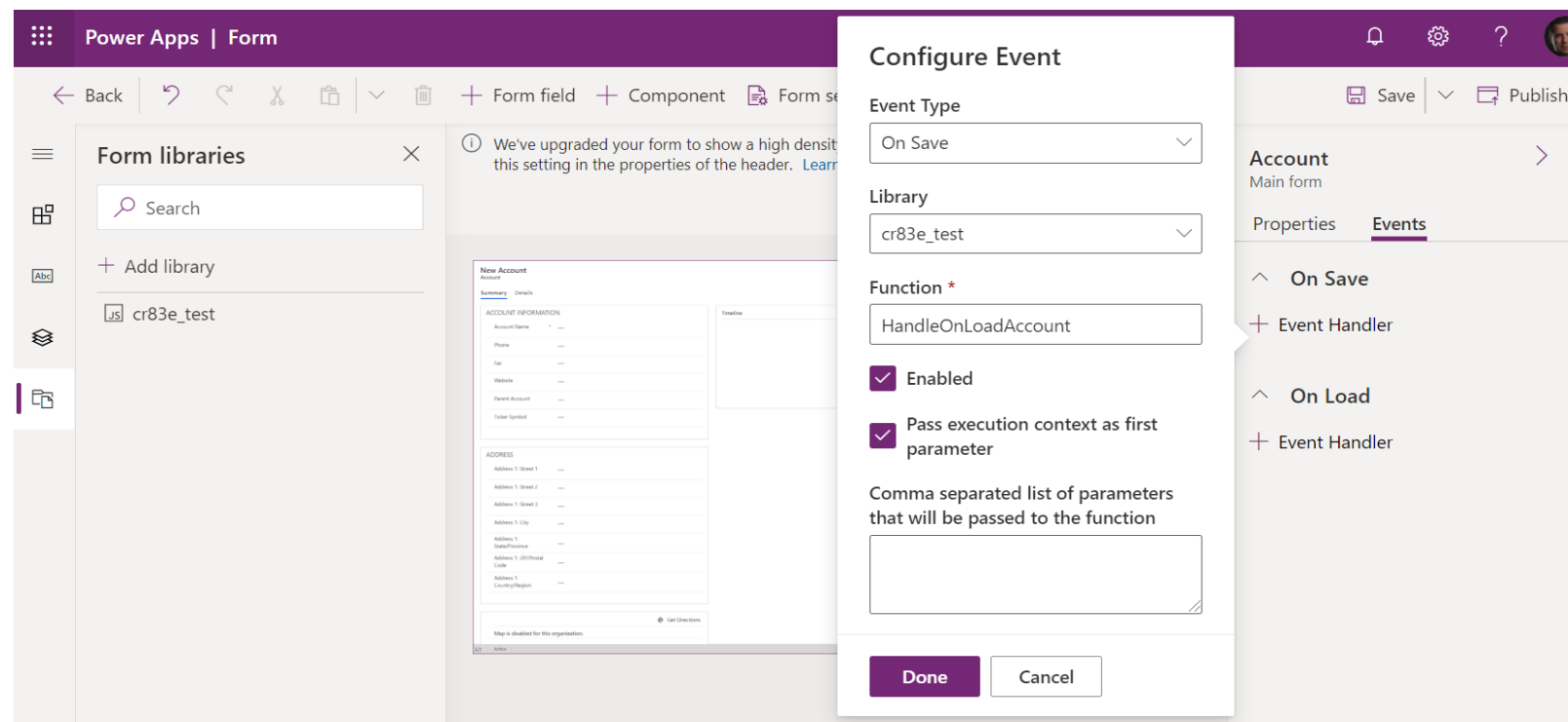


Registration is required

Registering via form properties

Registering event handlers using form properties creates a static configuration of event handlers at design time.

- **Form** – This allows you to register **OnLoad** and **OnSave** event handlers.
- **Columns** – This allows you to register an **OnChange** event handler if column data is changed



Give me some context

Execution context

When you register an event handler, you can have the execution context passed as the first parameter.

Account
Main form

Properties Events (1)

^ On Save
+ Event Handler

^ On Load
Table Column Dependencies
Select table columns

Handlers
LearnLab_handleAccountOnLoa ...
+ Event Handler

Configure Event

Event Type
On Load

Library
cr83e_RegHandler.js

Function *
LearnLab_handleAccountOnLoad

☒ Enabled

☒ Pass execution context as first parameter

Comma separated list of parameters that will be passed to the function

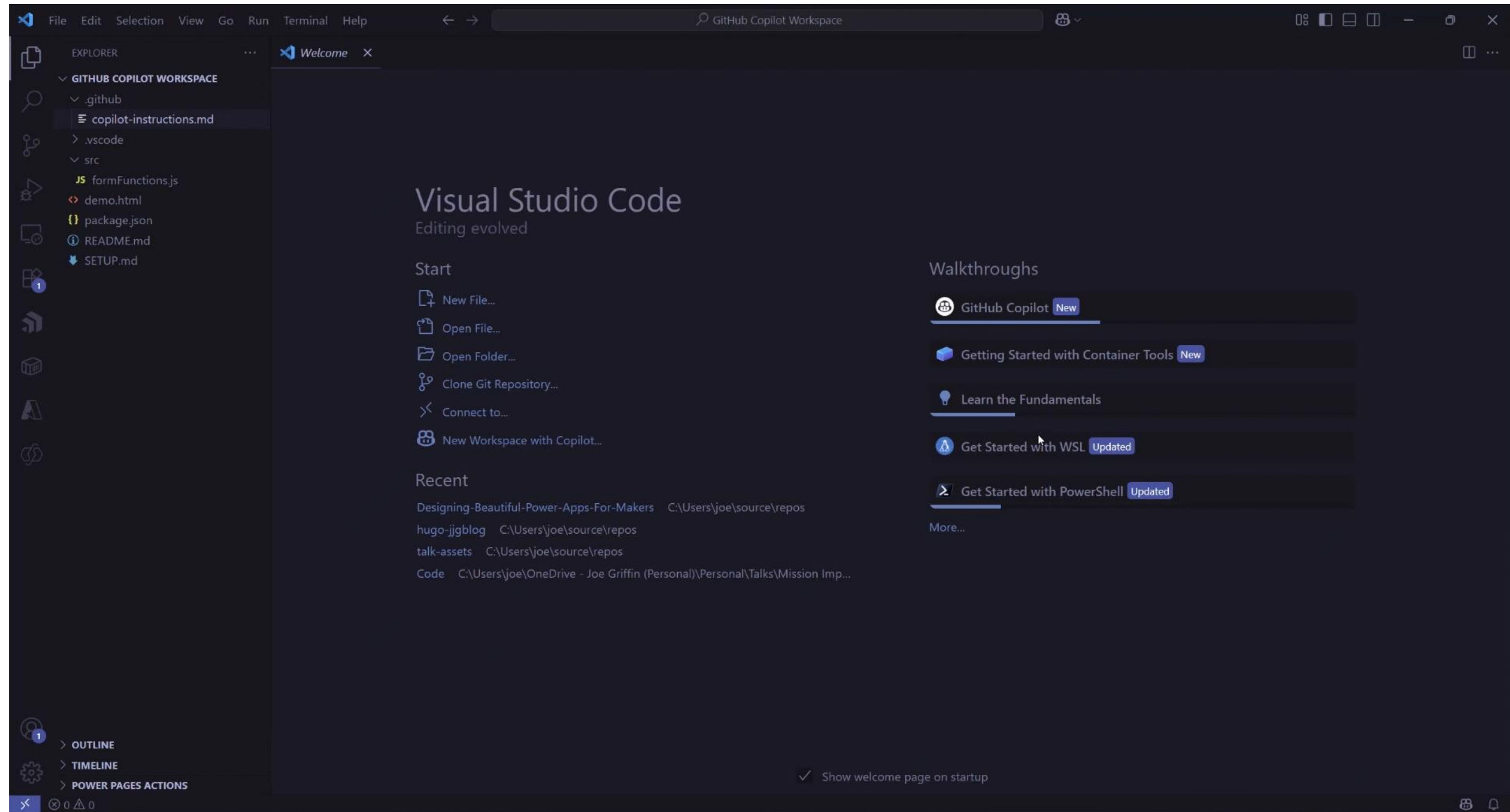
Done Cancel

```
1 function LearnLab_handleAccountOnLoad(executionContext)
2 {
3     //your logic goes here
4 }
```



Ohhh, we vibing now 🧐

Use GitHub Copilot to achieve your inner “vibe coder”



DEMO: Developing a simple form function to open a custom page



That's a fancy PCF you have there

Introduction to Power Apps component framework

Microsoft Power Apps component framework helps you create reusable components that can be used within your Power Apps applications.

The screenshot displays a Microsoft Dynamics 365 interface for an Opportunity record titled "4G Enabled Tablets". The top navigation bar includes standard actions like New, Refresh, Close as Won, Close as Lost, Recalculate Opportunity, Convert to Work Order, Process, Assign, Email a Link, Delete, Share, Follow, and Flow. The record header shows the Opportunity name, a dropdown menu, and key fields: Est. Close Date (3/6), Est. Revenue (\$3,257,500.00), and Status (In Progress). Below the header is a process bar for the "Opportunity Sales Process", which is active for 6 days and shows stages: Quality, Develop, Propose (6 D), and Close. The main content area is divided into three sections: Summary, Timeline, and Relationship Assistant. The Summary section lists details such as Contact (Vincent Lauriant), Account (Southridge Video), Purchase Timeframe (This Quarter), Currency (US Dollar), Budget Amount (\$6,922,000.00), Purchase Process (Individual), and Description. The Timeline section shows a list of recent activities, including "Task completed by Spencer Low (Sample Data) - Saturday 6:39 PM" and "Send Email - Follow up with relevant products". The Relationship Assistant section displays a list of stakeholders, including Vincent Lauriant.

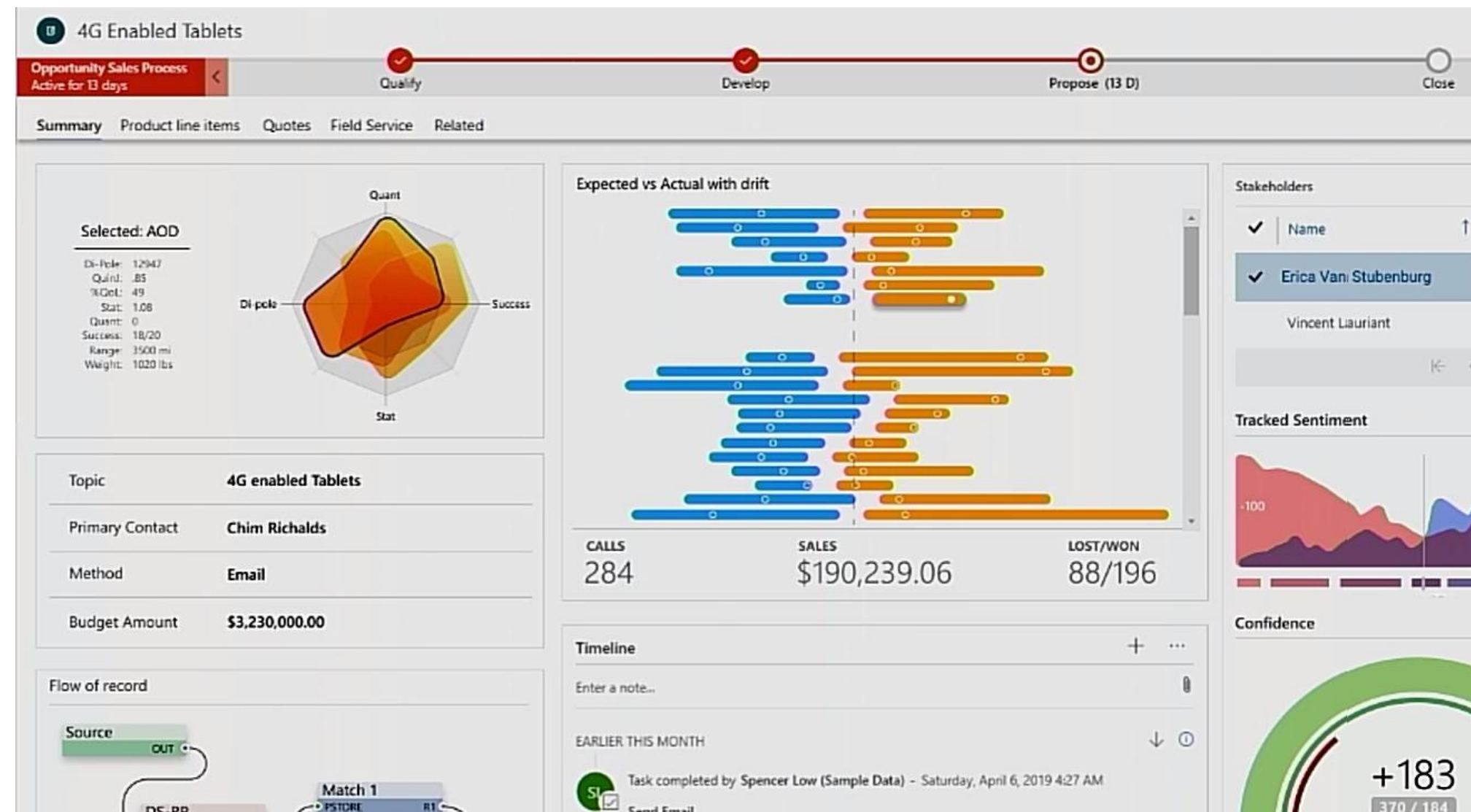
Topic	4G Enabled Tablets
Contact	Vincent Lauriant
Account	Southridge Video
Purchase Timeframe	This Quarter
Currency	US Dollar
Budget Amount	\$6,922,000.00
Purchase Process	Individual
Description	---



That's a fancy PCF you have there

Introduction to Power Apps component framework

However, if you reconfigured your app to use custom Power Apps components, your app might look something like the following image.



Break this down for me

What are PCF Code components?

1. Reusable rich user interface controls
2. Created by developers
3. Developed using HTML, CSS, and TypeScript
4. Packaged as solutions
5. Can be used in model-driven apps, canvas apps, custom pages and Power Pages sites



Check the defaults

Enable for canvas apps and custom pages

Power Apps component framework for canvas apps

Enables Power Apps component framework feature that allows the execution of code that may not be generated by Microsoft when a maker adds code components to an app. Make sure that the code component solution is from a trusted source. [Learn more](#) 

Allow publishing of canvas apps with code components



Delete disabled users

Show the "Delete user" option in "Settings > Users" for disabled users, so you can permanently delete them from this environment. [Learn more](#) 

Enable delete disabled users



TDS endpoint



Save

Cancel

<https://aka.ms/ppac>



Why should I care?

Power Apps code component framework advantages

A few of the advantages that you are afforded as a result are:

- Access to a rich set of framework APIs that expose capabilities like component lifecycle management, contextual data, and metadata
- Support of client frameworks such as React and AngularJS
- Seamless server access through Web API, utility and data formatting methods, device features like camera, location, and microphone, in addition to easy-to-invoke UX elements like dialogs, lookups, and full-page rendering
- Optimization for performance
- Reusability
- Use of responsive web design principles to provide an optimal viewing and interaction experience for any screen size, device, or orientation
- Ability to bundle all files into a single solution file



Choose your type

Types of components that you can add

1. **Field:** A custom control for a column on a form.
2. **Dataset:** A custom grid (or subgrid) to display data in a tabular format.
3. A component that displays content from external services



DEMO: Creating and deploying a simple code component





If only there was a community for this...

PCF Gallery

**PCF Gallery**

Store

Authors

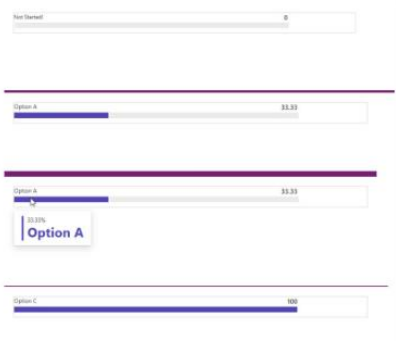
Ideas







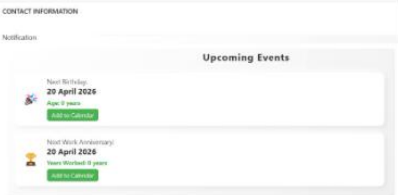
Submit a new PCF Control



Application Progress Bar

A control built to render a progress/stepper bar inside Dynamics 365 or Power Apps forms.





Birthday and Work Anniversary

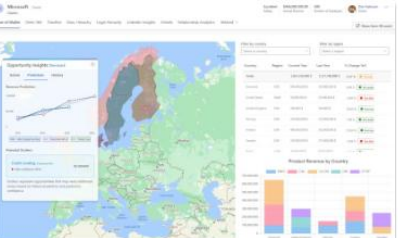
A control to show the next Birthday and Work Anniversary for a user, calculated from input dates.



BIRTHDAY

WORK

ANNIVERSARY



Share Of Wallet

A control to combine interactive map visualization with detailed opportunity analytics.



AZURE MAPS

WALLET

DATASET



Score Indicator

A control that displays a whole number value (0-100) using a modern Fluent UI React gauge chart.

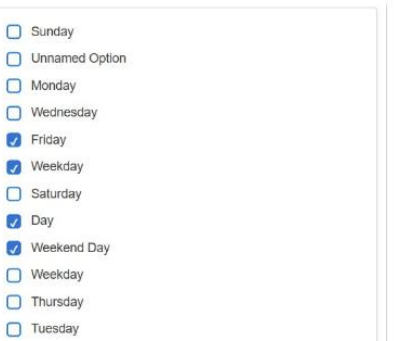


SCORE

INDICATOR

GAUGE

CHART



Multi Option Set Control

A control designed to allow users to select multiple records from a dataset.



DATASET

OPTIONSET



File Upload ★

The File Upload PCF component allows users to upload files directly within a form to Dataverse.



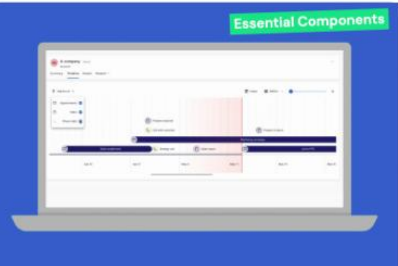
FILE UPLOAD





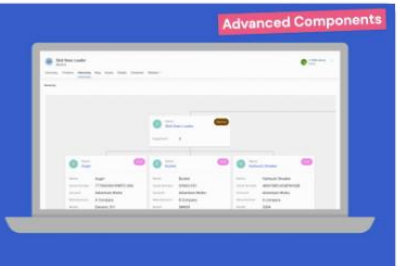
Scheduler

A control for event management, drag-and-

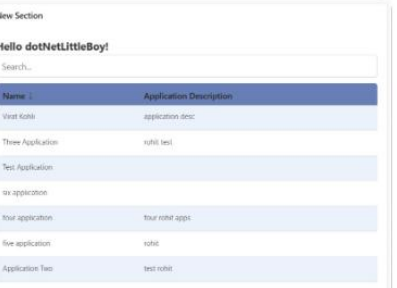


Visual Timeline ★

SummaryThe Timeline PCF component visually displays chronological events in a



Hierarchy Control with Tree View navigation (Multiple entities support) ★



ReactDetailsListSubgrid

A control built using React's DetailsList component to

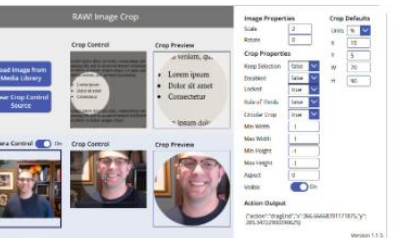
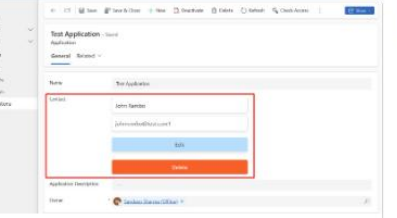


Image Cropper

A control to provides a modern, accessible, and



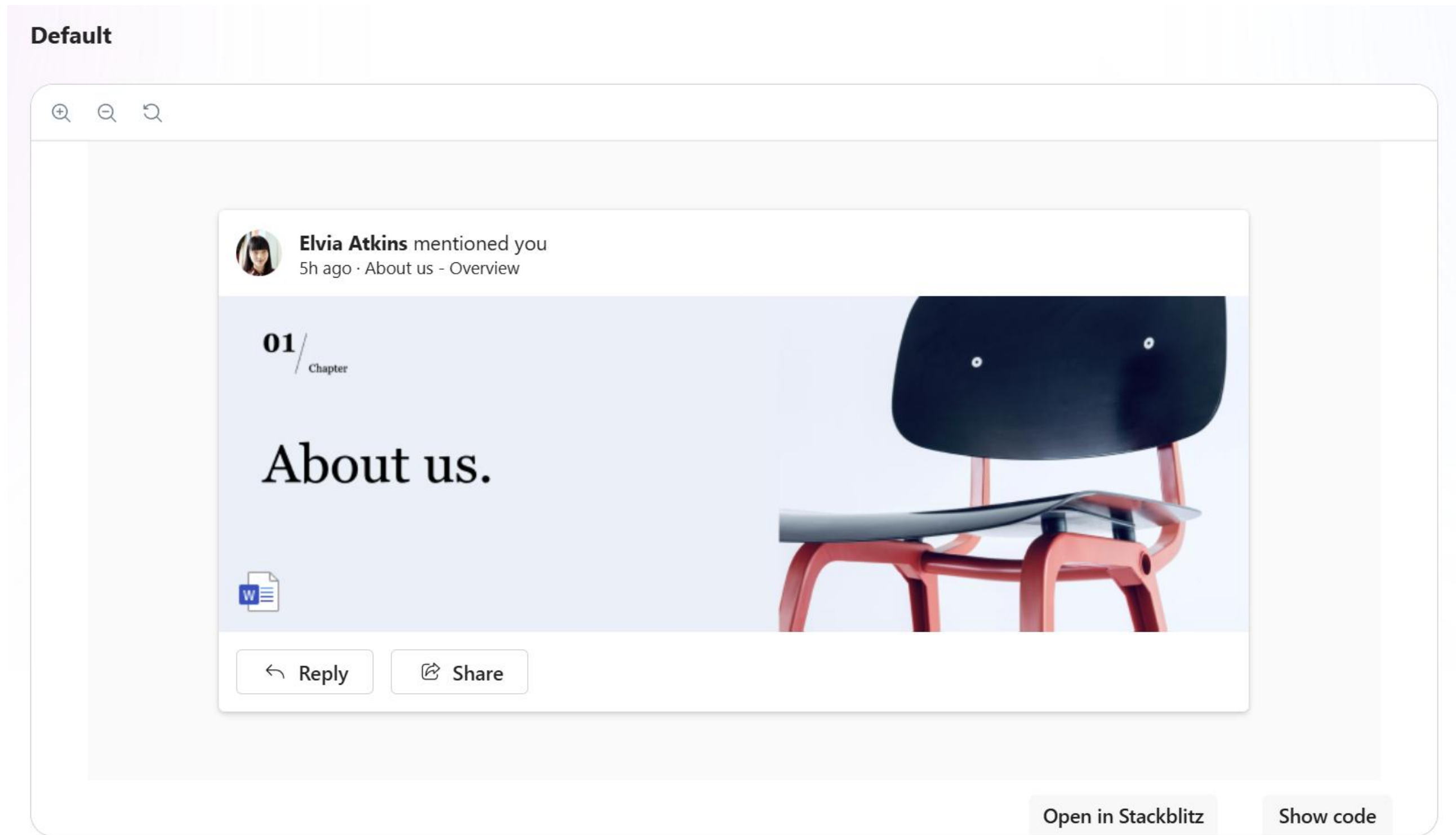
EnhanceLookupEdit

A control that enhances the standard lookup field by

<https://pcf.gallery>

Make your controls super fancy

Fluent UI + React + PCF Controls = Better Together 🤝



<https://react.fluentui.dev/?path=/docs/concepts-introduction--docs>



Help me, I'm indecisive

JavaScript versus TypeScript versus React

JavaScript	TypeScript	React
Used for client scripting	Used for client scripting and PCF control development	PCF control development only
Not strongly typed	Strongly typed	Strongly typed
Prone to runtime errors; additional testing / debugging often needed	Enforces compile time checks, reducing common errors.	Enforces compile time checks, reducing common errors.
Minimal support for ALM	Supports more complex ALM scenarios	Only contributes in ALM when used in PCF control development
Easy to learn and understand	More complex to learn and master	More complex to learn and master





Lab 1 – CREATE MODEL-DRIVEN POWER APP

In this lab, you will do the following:

- Create a new table called Purchase Order and customise it accordingly.
- Add in the Account and Contact tables to your solution and customise accordingly.
- Create and define the configuration for your model-driven app, before testing it further.
- Add some test records into the system.
- Configure the integration with the backend ERP API.
- Create a cloud flow to generate purchase order request documents, and have this available to trigger from the model-driven app.

Expected completion time: 60 minutes

API Key - Please check the spreadsheet: <https://bit.ly/BeautifulPowerApps-Credentials>



DEMO: Power Apps types



Module 3: Working with Custom Pages



Model-driven apps + Custom Pages

Increase user experience and **business productivity**

Bridge functional gaps

Merging **structure**, custom logic and design

Extending standard capabilities

Connect to external data sources and access **organisational data**

User-centric development

Display as a side pane, full page or a dialog

Fusing low-code and pro-code

Developed using Power Fx – Enabled by JavaScript – Supports custom code components (PCFs)



It's all about the use cases

Common scenarios where custom pages are **excellent**

Landing zones

Main screen and dashboard

Custom UI for business processes

View PDFs, start approval processes and display record information

Integrations

Connect to other data sources than Dataverse. 1400+ connectors.
Trigger Power Automate.

Routing

Optimize the user journey. Open a specific form, redirect to another page and launch websites



Custom Pages Complexity

Cool designs might meet high difficulty

Unexpected behaviour

“It doesn’t look like that for me”

Use solutions, be careful with preview features and test *a lot*. DateTime values, language & disappearing controls

Build for performance

Life is too short to wait for unsorted data

Use low-code solutions where low-code is **GREAT!**

Increased difficulty

Adding another layer of complexity with **JavaScript**

Maintenance, web resources and new requirements

User authentication & permissions

Custom pages utilizes a shared connection infrastructure

User consent (connector authentication) spans across **EVERY** custom page in the MDA.



Data – lazy loading

You don't need 1000 records in the app at the same time...

Delegation

What ever that means

- Off load data processing tasks to the server side
- **Large** datasets – more than **500-2000** records
- Data accuracy – use Filter(), Sort(), StartsWith()
- Locally processed – Collect(), ClearCollect(), Search()



Design considerations

One size **doesn't fit all**

Build responsive apps

Your end-users will thank you

- Utilise containers to align and group controls
- Automatic **resizing** of screen elements
- **Parent.Height** & **Parent.Width**
- X, Y, Height and Width – Position and dynamic values

Consistency

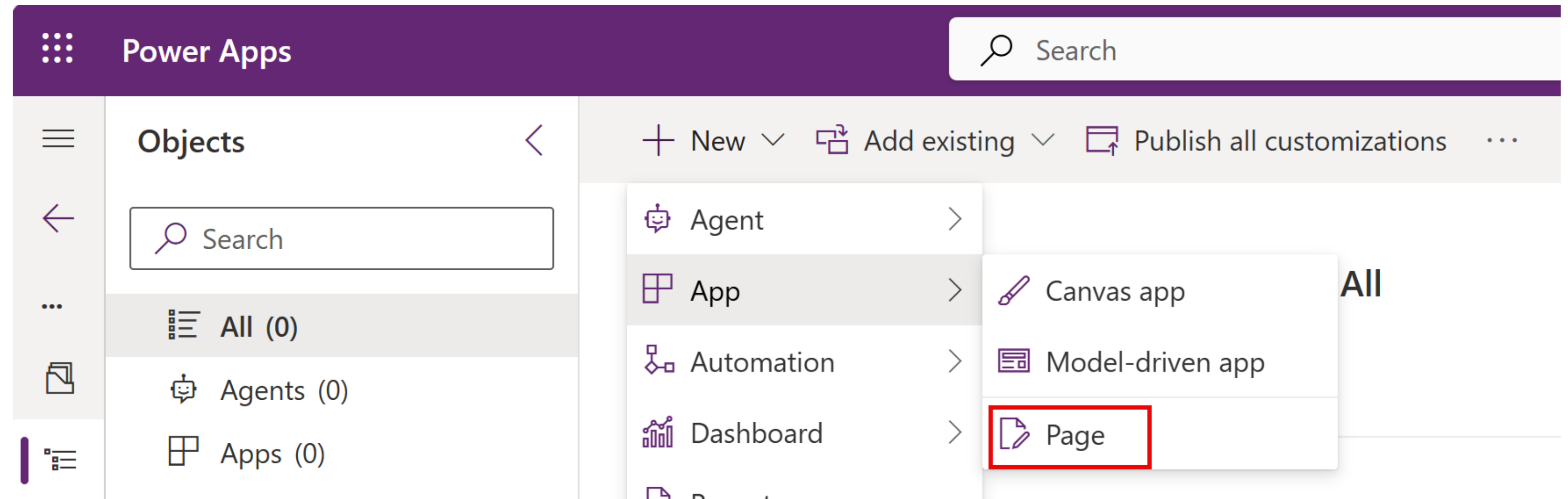
Make it look sexy

- Reuse components and YAML
- Color scheme, fonts and company design
- Not every control is **supported** (i.e PDF viewer, barcode scanning, attaching files and taking pictures)



Show me it in action! 🙄

Creating a custom page



- Multiple ways to create a custom page
 - Solution
 - Model-driven app designer



Soooo many options

Settings and configuration

- Name your page
- Provide a description
- Set an icon or image
(brand logo)
- Set background fill
- This is what shows up
when the page is
loading

Settings
General
Display
Copilot
Updates
Support

General

Name

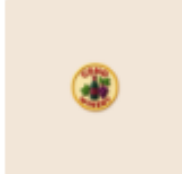
Coho Winery Landing Page 

Description

Provide a description to help end-users find and use your app.

Landing page for Coho Winery Purchase Order Application

App icon



Icon

Select icon 

 Add image

Icon background fill

 #f4e6d7

Icon fill



Close



Just tell me how to do it!

Designing a custom page

Layouts Templates

These layouts adapt to different screen sizes. When you preview the app, you can choose a different screen size and see how it responds.



Split screen



Sidebar



Header and footer

- Responsive design with containers
 - Vertical containers
 - Horizontal containers
- Set properties of containers to handle flexible widths and heights.
- Layouts available when creating a custom page.
- Subset of canvas controls supported.
- Theme set to match theme of model-driven app.



Custom Page embedded

Adding a custom page to an app

- Add custom page
- Add to navigation
- Edit subarea to set the title used in navigation and privileges to access the page
- Publish the app

← New page

Select one to start with

☐ Create a new custom page

☒ Use an existing custom page

☒ Contacts custom page
 ☐ Dataverse Actions Page
 ☐ Overview

☒ Show in navigation

Design and build the page you want by dragging interactive components onto the canvas.

Add Cancel



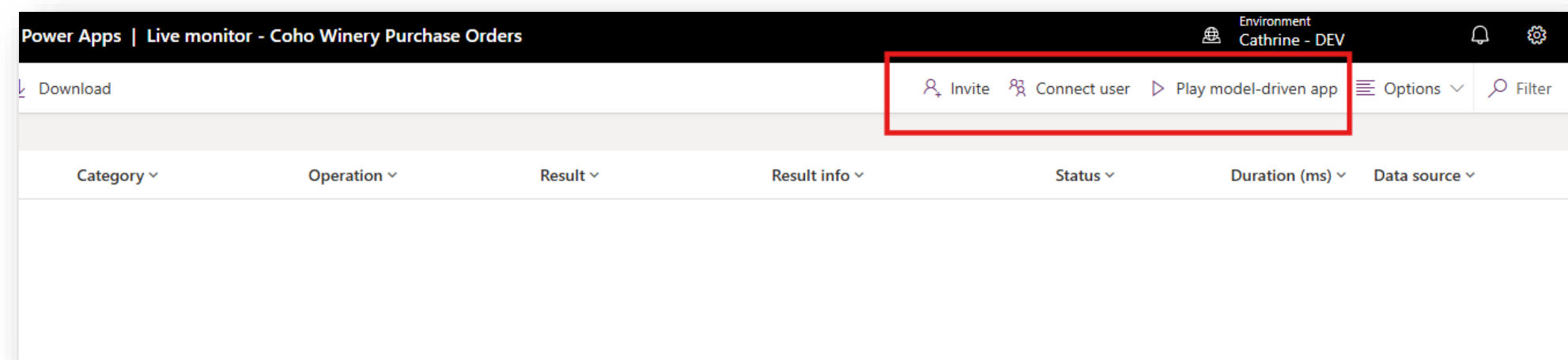
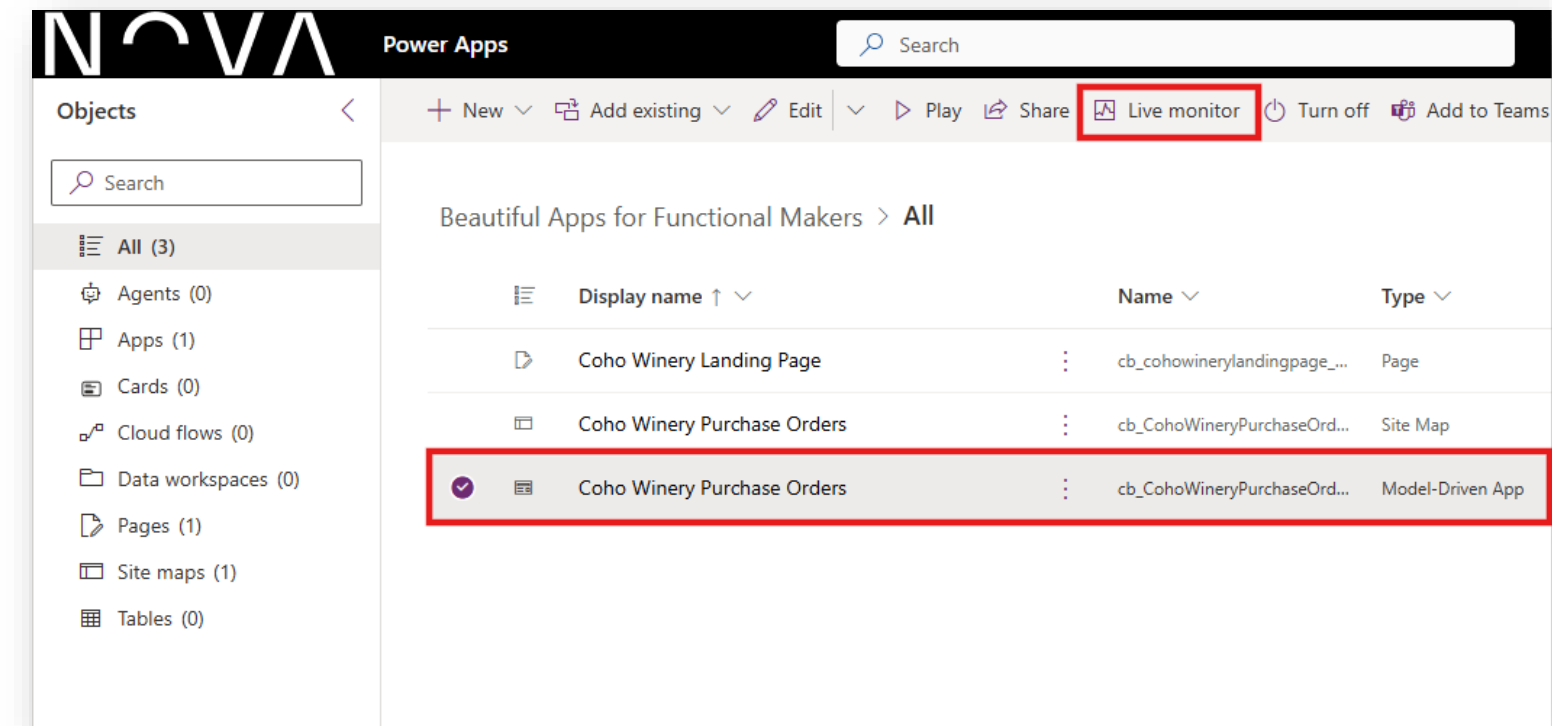
DEMO: Getting started with Custom Pages



I See You 🧐

Monitoring

- In the solution, select the MDA and Live monitor
- The monitor will open. Choose to play the app or connect another user to see their details



It's an infestation

Debug and improve

Model-driven app session					
Id	Time	Category	Operation	Result	Result info
59	12:57:25.477	Network	Fetch	Success	
60	12:57:25.477	Network	Fetch	Success	
61	12:57:25.478	Network	Fetch	Success	
62	12:57:25.485	Network	Fetch	Success	
63	12:57:25.490	Network	XMLHttpRequest	Success	
64	12:57:25.505	Network	AssetLoad.IFrame	Success	
65	12:57:25.553	Network	Fetch	Success	
66	12:57:25.605	Network	XMLHttpRequest	Success	
67	12:57:25.676	Network	XMLHttpRequest	Success	
68	12:57:25.777	Network	Fetch	Success	
69	12:57:25.778	Network	Fetch	Success	
70	12:57:25.779	Network	Fetch	Success	
71	12:57:25.693	KPI	PageNavigationModuleRe...	Success	
72	12:57:26.356	ClientAPIChecker	XrmlInternal.Copilot	Success	
73	12:57:26.585	ClientAPIChecker	XrmlInternal.Copilot	Success	
74	12:57:26.084	Network	XMLHttpRequest	Success	
75	12:57:26.167	Network	AssetLoad.Script	Success	
76	12:57:26.169	Network	XMLHttpRequest	Success	
77	12:57:26.187	Network	Fetch	Success	
78	12:57:26.551	Network	AssetLoad.IFrame	Success	
79	12:57:26.592	Network	Fetch	Success	
80	12:57:26.651	Network	Fetch	Error	Bad Rec
81	12:57:26.670	Network	Fetch	Success	
82	12:57:26.699	Network	Fetch	Success	

- In the **Live monitor** everything the user selects will show up
- Use the information to improve the performance and issues in the app



Please contain your excitement

Container properties

Properties

CONTAINER ?

cntMainVertical

Display Advanced

Direction

Vertical

Justify (vertical)

Align (horizontal)

Gap

16

Horizontal Overflow

Hide

Vertical Overflow

Hide

Wrap

Off

Position

34.15

19.2

X

Y

Size

1297.7

729.6

Width

Height

Padding

16

16

Top

Bottom

16

16

Left

Right

Color

Border

0

Border radius

4

Drop shadow

None

Visible

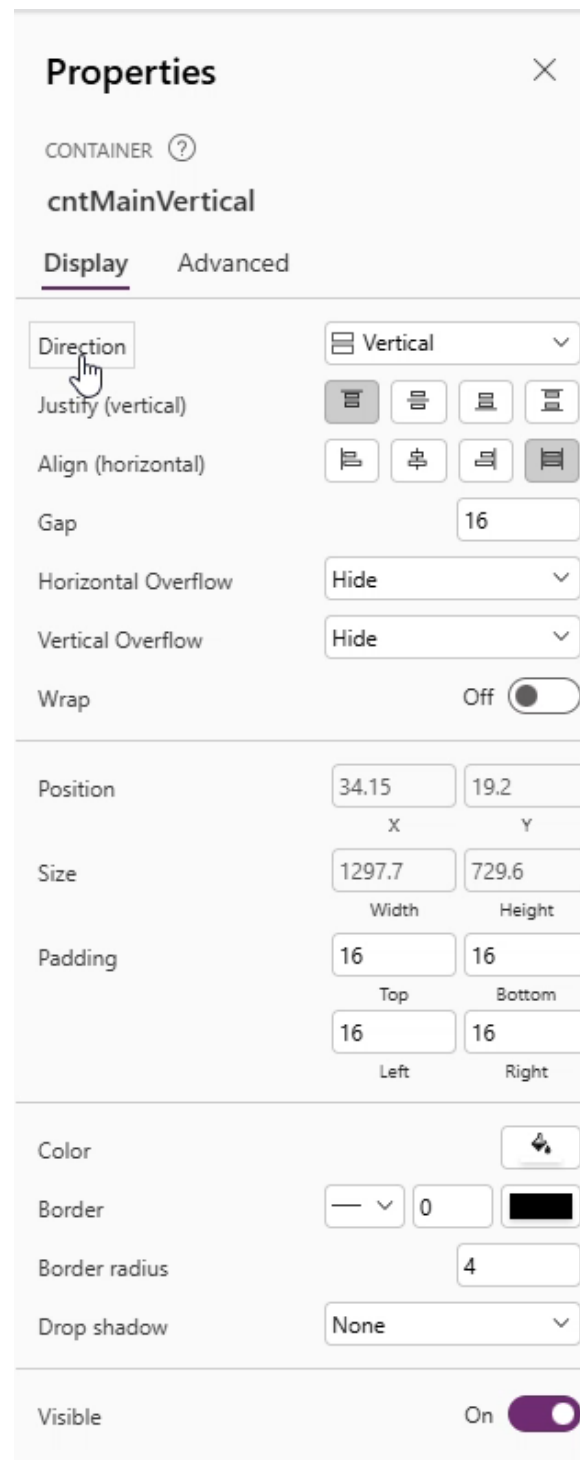
On

- Properties that will take you some time to get used to and master.
- Make sure to test, set values, and play app with different sizes
- Control alignment in containers; no manual control, strict rules for item placement
- Setting the parameters for adjustments and dynamic values, not hardcoded; avoid using drag and drop



Lemme give you a tip there

Container tips



- If you need the container to expand with dynamic text, turn **Flexible height = On** and keep **Vertical Overflow = Hide**.
- If you see unexpected scrollbars, check **Wrap** first; wrapping usually removes the need for horizontal scrolling – Text shows as cath.... instead of cath
rine
- To get equal column widths, give sibling containers **Fill portions = 1** (or 2, 3...) and set **Align in container = Stretch**.





Less is more, people 🙄

Styling containers

- **Fill (Color – Fill)**

- Background color of the container – can be transparent if you have a background color

- **Border Color / Border Thickness / Border style**

- Styling properties for borders – try and fail.
Usually invisible for main container, but can add depth and highlight content if needed

- **Border Radius**

- Border roundness of corners – often changed
border radius between 4-10

- **Drop Shadow**

- Shadow intensity – Light is my go-to

PURCHASE ORDERS
ERP data in model-driven apps

Search for vendor or Purchase Order Number... Show orders from date 31.12.2001 Number of records 25 Search Reset

PO number	Datasource	Vendor	ID	Edited	Processed	Status	Warehouse	Lines	Records: 15
PO12345	Dataverse	Vendor a	V001	12.09.2025	false	Pending	WH01	10	>
PO12346	Dataverse	Vendor b	V002	09.09.2025	false	Approved	WH02	5	>
PO12347	Sharepoint	Vendor c	V003	04.09.2025	true	Rejected	WH03	15	>
PO12348	Dataverse	Vendor d	V004	13.09.2025	false	Pending	WH04	20	>
PO12349	Sharepoint	Vendor e	V005	14.09.2025	true	Approved	WH05	8	>
PO12345	Dataverse	Acme corp	Vendor001	09.09.2025	false	Success	WH001	10	>
PO67890	Sharepoint	Global supplies	Vendor002	11.09.2025	false	Pending	WH002	5	>
PO11223	Dataverse	Tech distributors	Vendor003	07.09.2025	false	Failed	WH003	15	>
PO44556	Sharepoint	Fasttrack logistics	Vendor004	13.09.2025	false	Success	WH004	8	>
PO77889	Dataverse	Northern lights	Vendor005	12.09.2025	false	Pending	WH005	20	>
PO33445	Dataverse	Nextgen supplies	Vendor006	08.09.2025	false	Success	WH006	12	>
PO55667	Sharepoint	Alpha distributors	Vendor007	10.09.2025	false	Failed	WH007	6	>
PO88990	Dataverse	Global traders	Vendor008	05.09.2025	false	Pending	WH008	9	>
PO99887	Sharepoint	Pro logistics	Vendor009	06.09.2025	false	Success	WH009	11	>
PO12399	Dataverse	Blue horizons	Vendor010	04.09.2025	false	Failed	WH010	4	>

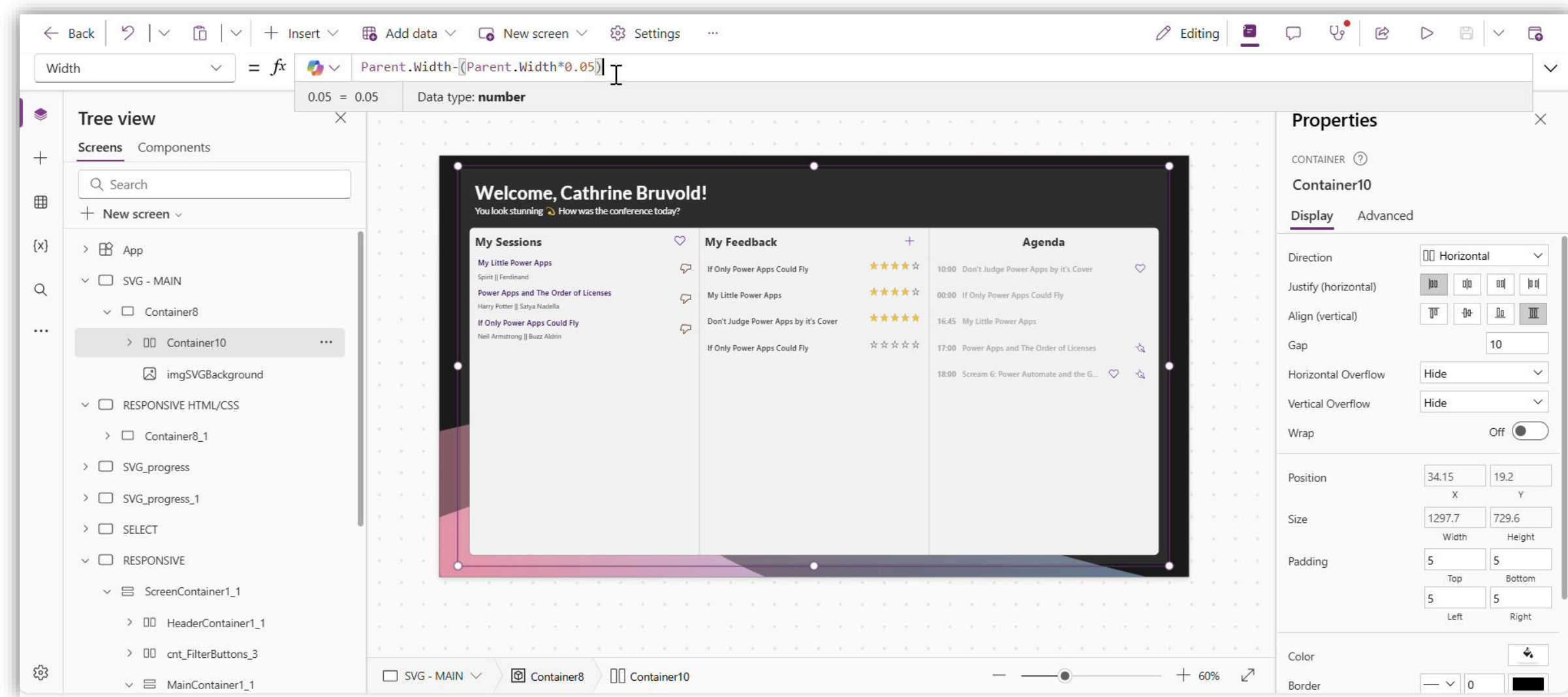
Building From Scratch 🤖

Takes foreverrrrrrr so reuse it

Container within container within container - Horizontal, vertical and container

App Width and **Height** are the two properties corresponding to the size of the screen.

Dynamically set width, height and position of controls based on percentage, the **Parent./Self.** Functions and formulas.



Pivot. PIVOT!!!

Positioning and sizing containers

- **Layering and dynamic values**
 - Use **Parent.** Values to automatically scale controls relative to the parent object
- **X, Y, Height and Width**
 - Main properties of controls used to dynamically size elements and position them. When the numbers are grey, a formula is used to set the values. DO NOT MANUALLY ADJUST 😊
- **Flexible height**
 - **On** when you have dynamic text values (will ignore Height/Width property), **Off** when you want to control the size. For instance, main containers that should have a set height.

After you create formulas, be careful not to adjust the elements manually.

Power Fx (center align)

X: $(\text{Parent.Width} - \text{Self.Width}) / 2$

Y: $(\text{Parent.Height} - \text{Self.Height}) / 2$

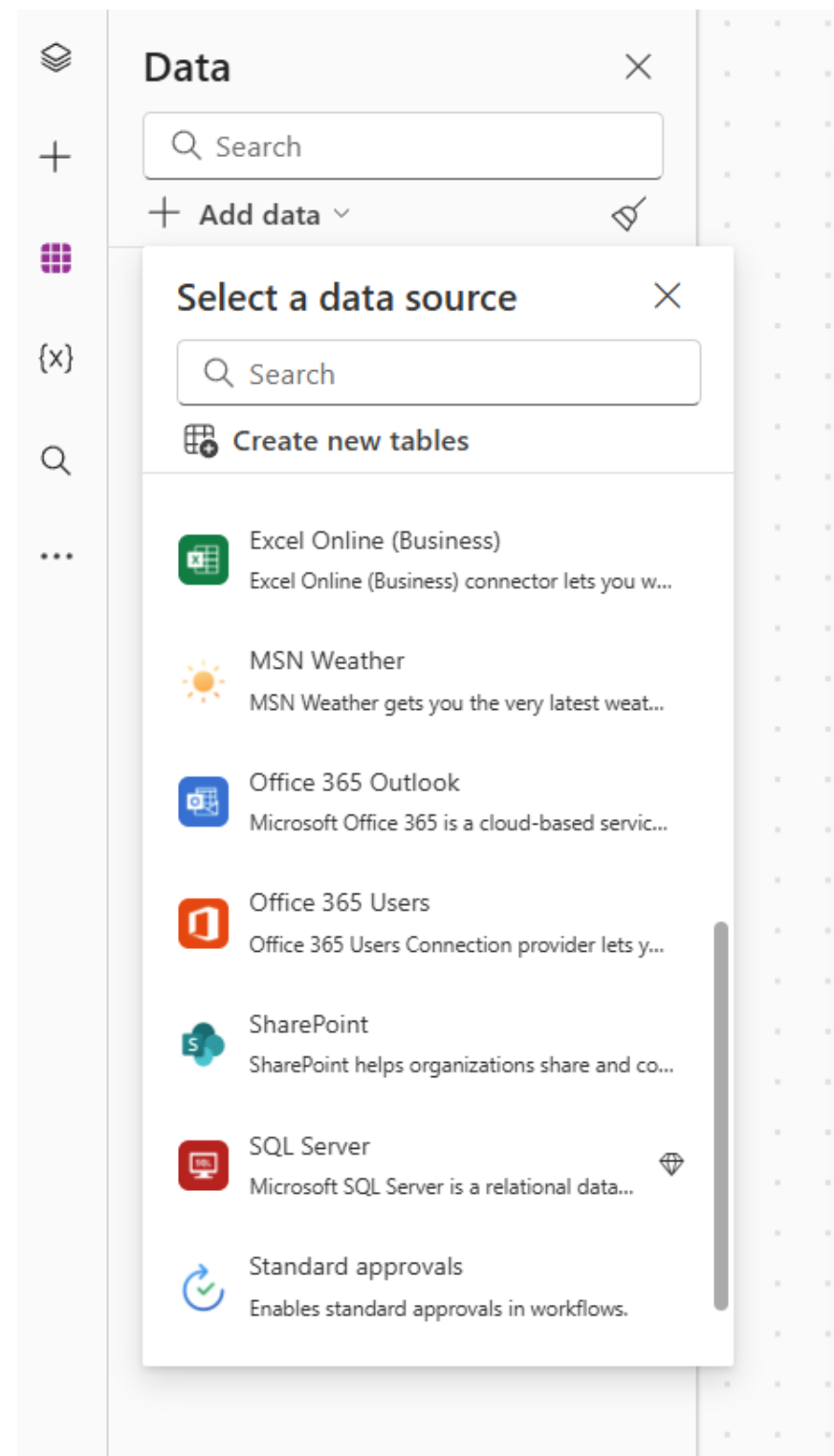
Width: $\text{Parent.Width} - (\text{Parent.Width} * 0.05)$

Height: $\text{Parent.Height} - (\text{Parent.Height} * 0.05)$



Connect to 1,400+ systems? Why not?

Working with connectors in custom pages



- Full access to both tabular and action-based connectors.
- Consider how data sources are shared and potentially accessed by users
 - Security
 - Fix connections
- Recommendations:
 - Limit number of connectors to 10



Named Formulas

Write reusable code once

Read-only global variables

Automatic recalculation of values.
No **timing-dependency** or manual
variable update required.



Benefits

Define once and reuse, **avoid redundant code** and utilise
lazy-loading principles

Code definition

App.Formulas: define formula name and logic to be
executed as **nfFormulaName** = expression;
Reference the **Named Formula** instead of code
duplication

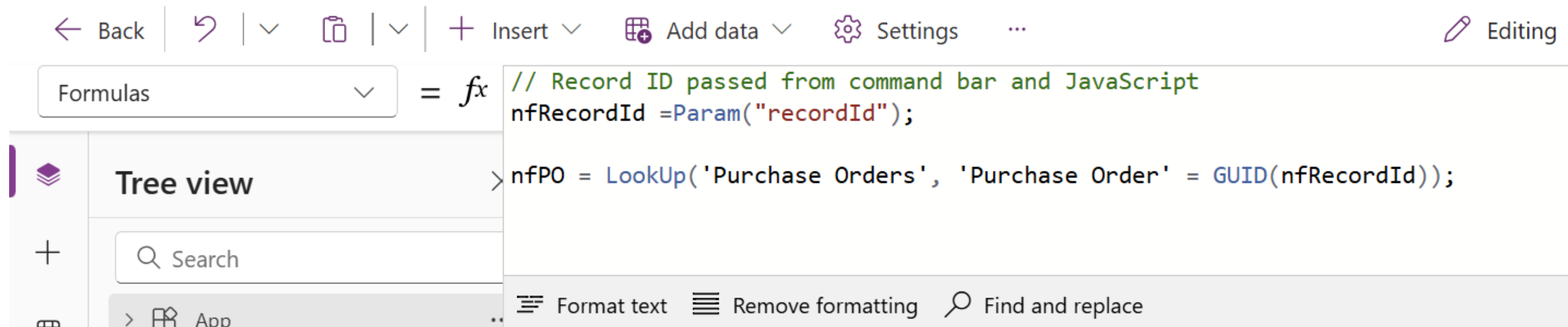


Named Formulas

Reduce the amount and **repetition** of code

Setting the Named Formula on **App.Formulas** containing the code you want to execute

Calling the Named Formula instead of the entire syntax every time



Named Formulas

App.Formulas

nfNamedFormula = **NAME DEFINITION**

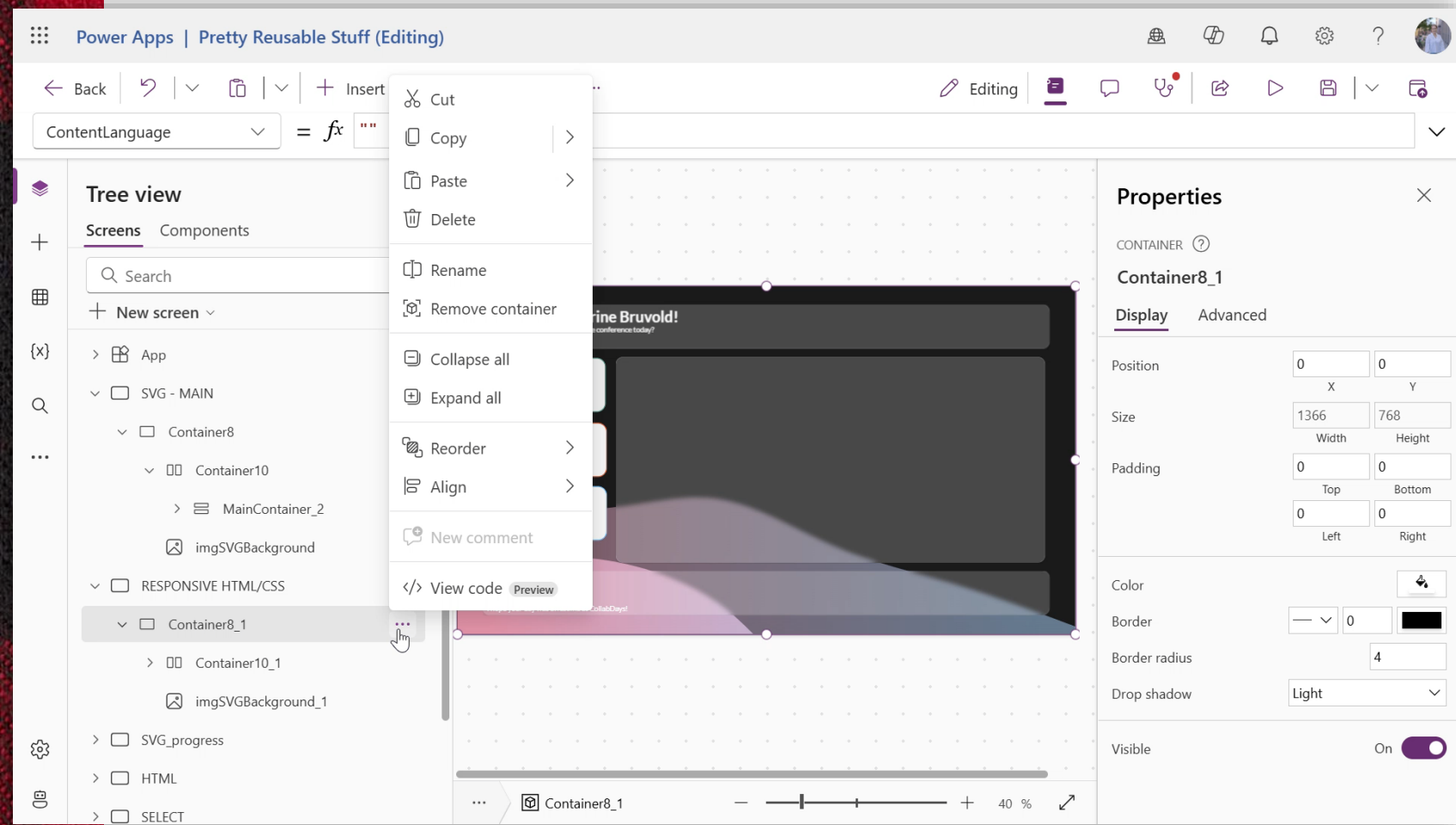
ShowColumns(
LookUp(
'Data source',
'Column name' = **EXPRESSION**
SetVariableValue
),
'Field 1',
'Field 2',
'Field 3'
);

Control property OnSelect

✓ **ClearCollect(** **REFERENCE**
colCollectionName,
nfNamedFormula
)

✗ **ClearCollect(colCollectionName,**
ShowColumns(
LookUp(
'Data source',
'Column name' = SetVariableValue
),
'Field 1',
'Field 2',
'Field 3'
))





Canvas YAML

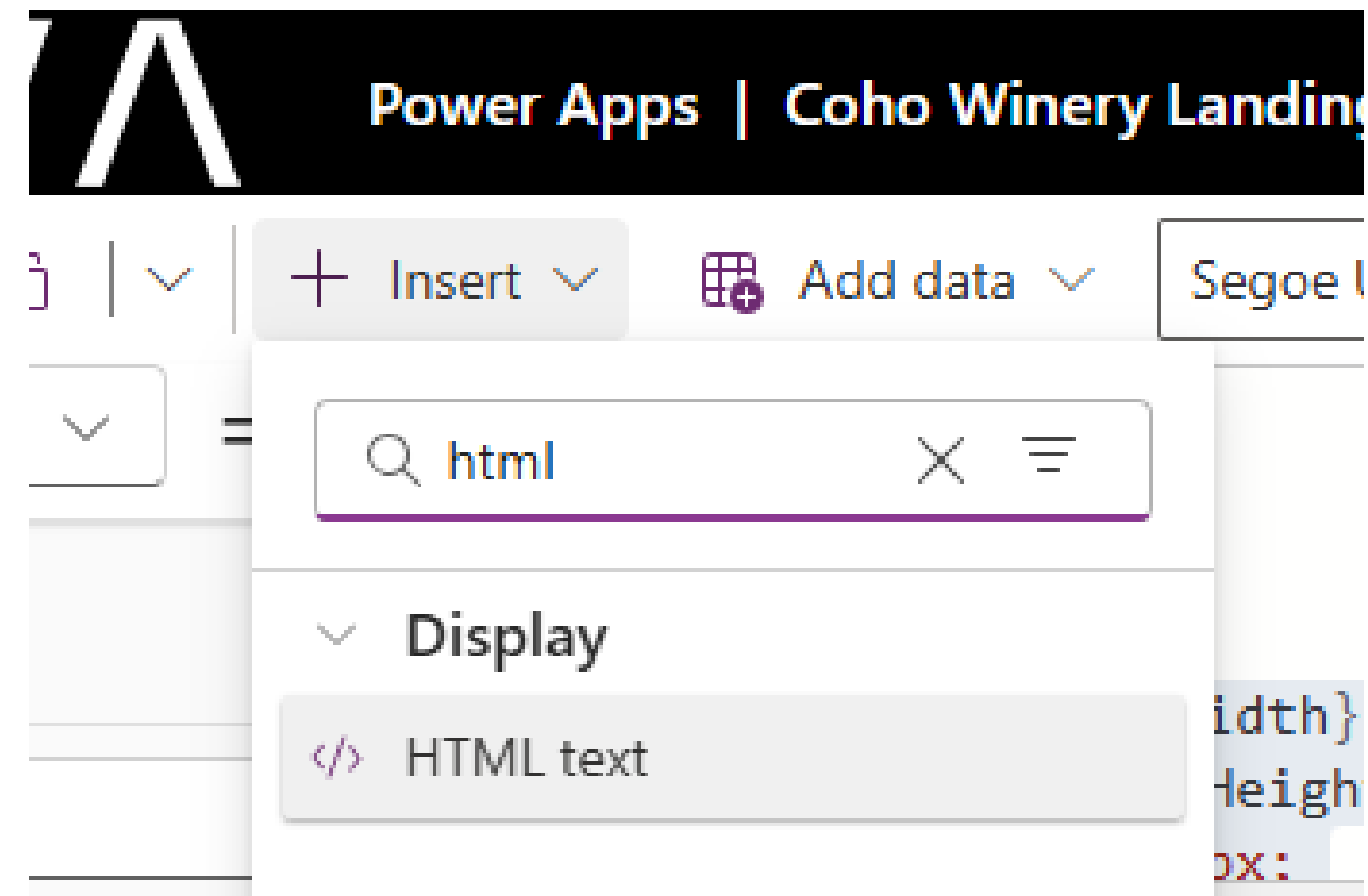
- Store and reuse YAML snippets to build apps faster
- Copy and paste YAML code from canvas apps controls and containers
- Use YAML to investigate the code and get a full overview of functions used, naming of controls and where they are located
- Increase reusability and save your favorite components in Visual Studio Code



App makeover time

HTML in Power Apps

- Build apps with more depth, customised design and advanced UI
- Use the HTML text control
- Add blur, colours, and custom “code”



Glass morphism

Example of HTML/CSS code snippet used to create Glass morphism features

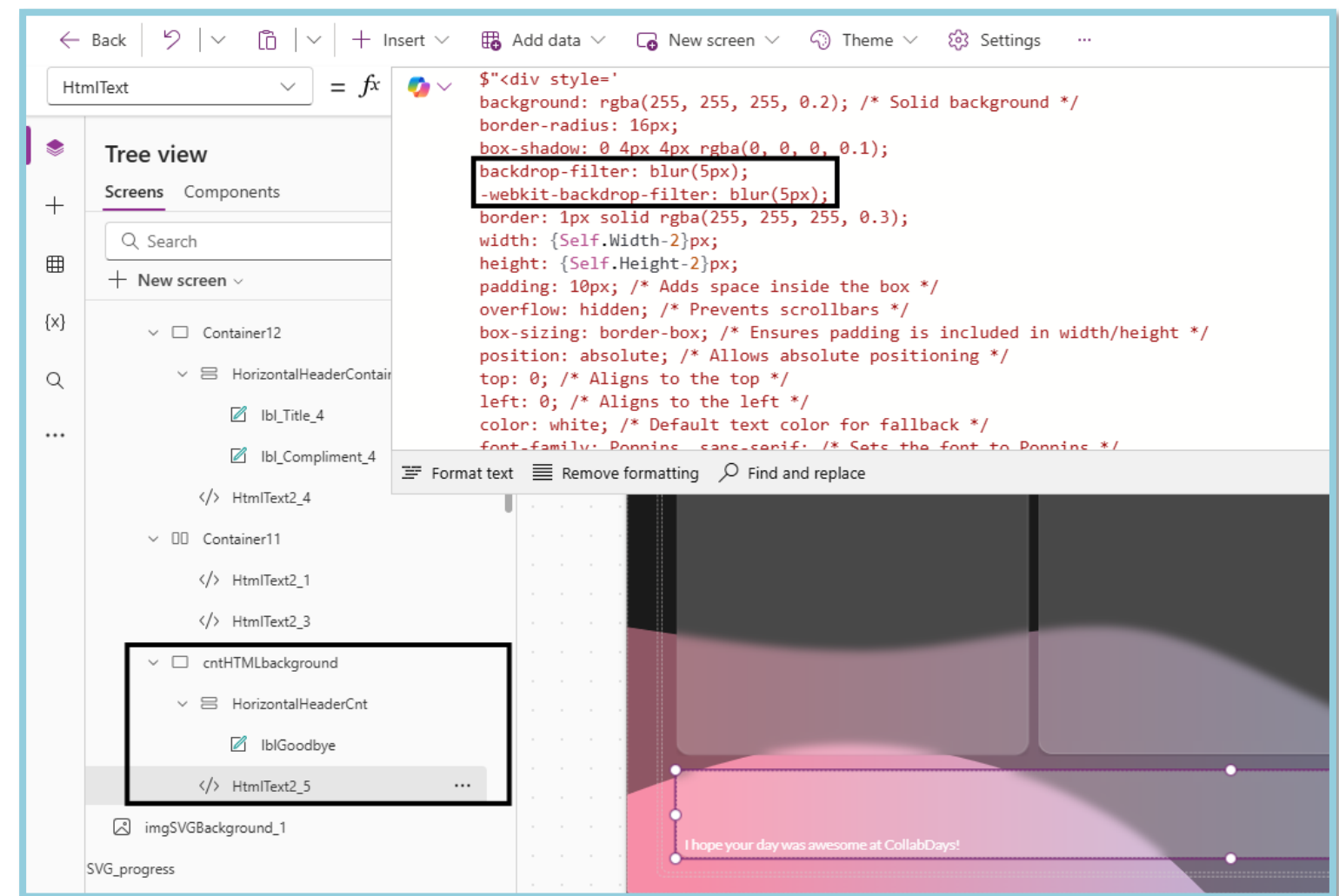
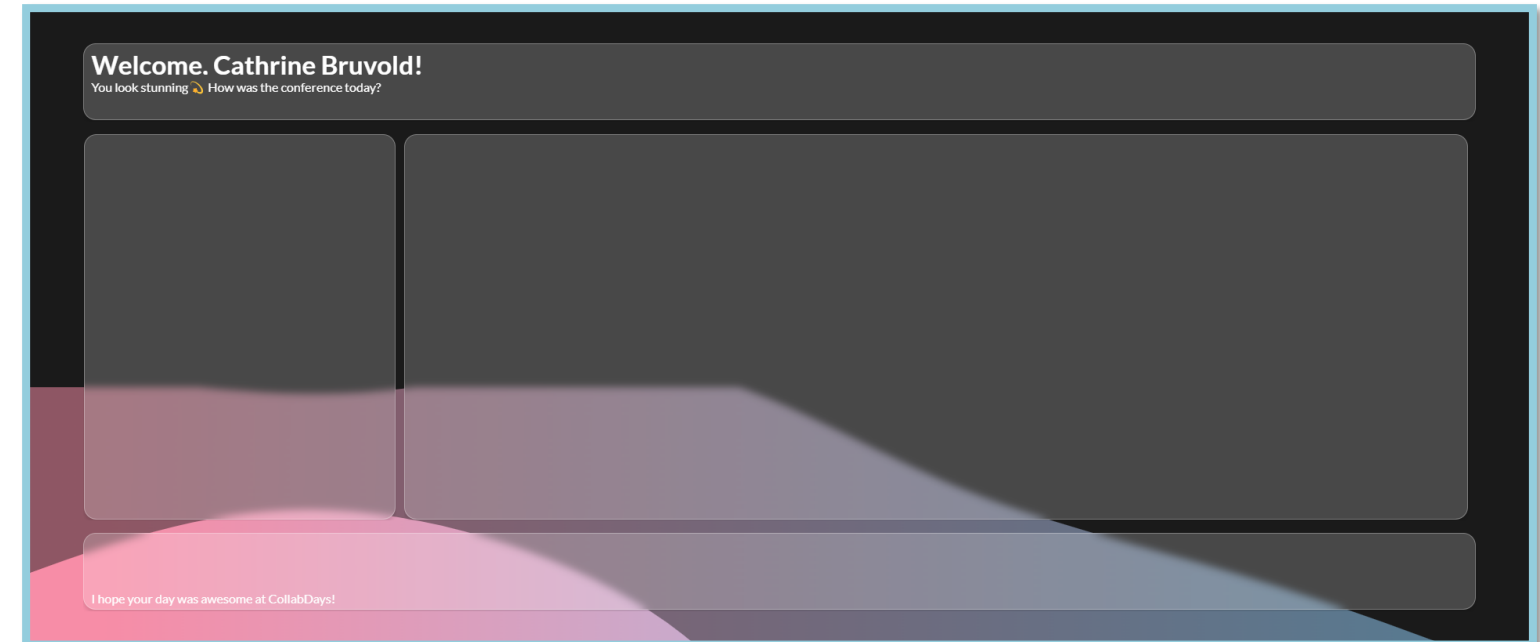
Create simple, yet effective depth to your canvas apps

Add containers to hold the HTML TEXT as a background, adjust width and height to **parent.Width** & **parent.Height** (responsive).

Add Horizontal or vertical containers above the HTML TEXT control

CSS generator: <https://css.glass/>

Get inspiration from kristine Kolodziejski's [YouTube](#)

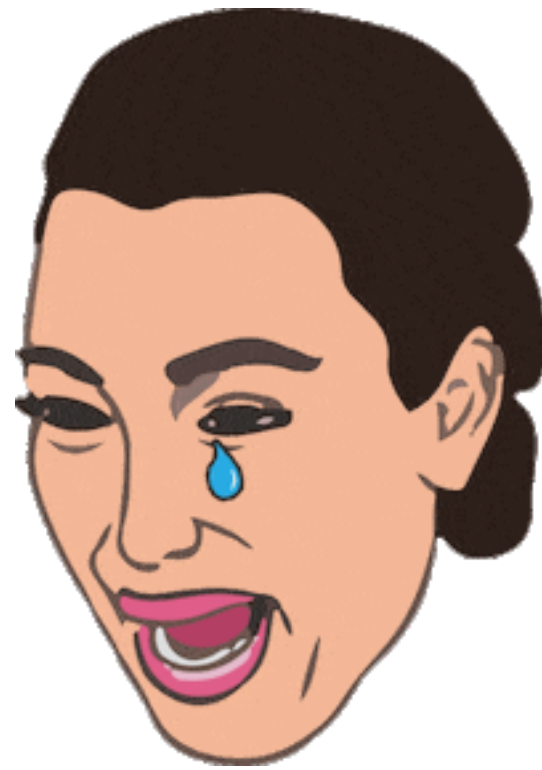


Demo: Advanced custom page development



Custom page with external data

Both pretty aaaaaand functional



“I just wanted to add some shading to my buttons.

Now I’m debugging regex validation rules and syncing data from an on-prem ERP system”



Custom page in MDA

Custom pages – Example Scenarios

Custom dialogs:

- Ribbon Commands
- Form events (**OnChange**,
OnLoad & **OnSave**)
- Side pane, dialog or full page

The image displays two screenshots of a custom MDA (Model-Driven App) page. The left screenshot shows a 'Purchase Order' form with a ribbon at the top containing 'Save', 'Purchase Order Info', and 'Share'. The form has tabs for 'General' and 'Related'. The 'General' tab is active, showing fields for 'Purchase Order Number' (1234567) and 'Owner' (Cathrine Bruvold). A 'Form assist' button is visible. The right screenshot shows a 'Purchase Order Confirmations' page with a 'Get Purchase Orders' button and a section for 'Purchase Order PDFs' displaying the number 1234567.



What sorcery is this???

Using JavaScript for custom pages

1. Start building

Create a solution for the custom page, model-driven app and the table where the page will be triggered from. Save the **logical name** of the custom page.

2. Script

Create a .js file with a function for triggering the custom page Dialog/Side Pane. Use the **logical name** of the custom page as a reference in the script as a parameter.

3. Configure the model-driven app

Add the .js file as a web resource in your solution. Open the **model-driven app designer**, add the custom page and configure the command bar on a Form or the Main grid to reference the web resource.

4. Test & iterate

Trigger the custom page and see what it looks like. Edit the custom page and JavaScript according to your needs (remember to publish)





Lab 2 – CREATE CUSTOM PAGES

In this lab, you will do the following:

- Build Custom Pages to be used as a full page, side pane and dialog in a Model-Driven App
- Connect to relevant data sources
- Fetch record details using the Param() function and connecting to the record context
- Develop responsive apps

Expected completion time: 60 minutes





Lab 3 – INTEGRATE CUSTOM PAGES IN MODEL-DRIVEN POWER APPS

In this lab, you will do the following:

- Embed the custom pages you built in Lab 2 into the model-driven app.
- Preparing the JavaScript web resource that will open the side pane from a command bar button on the Purchase Order form.
- Configure the command bar button to the Purchase Order form to open the side pane and pass the current record's ID.
- (Optional) Add JavaScript to the **OnLoad** event of the Purchase Order form to automatically open the side pane when the form loads.

Expected completion time: 60 minutes



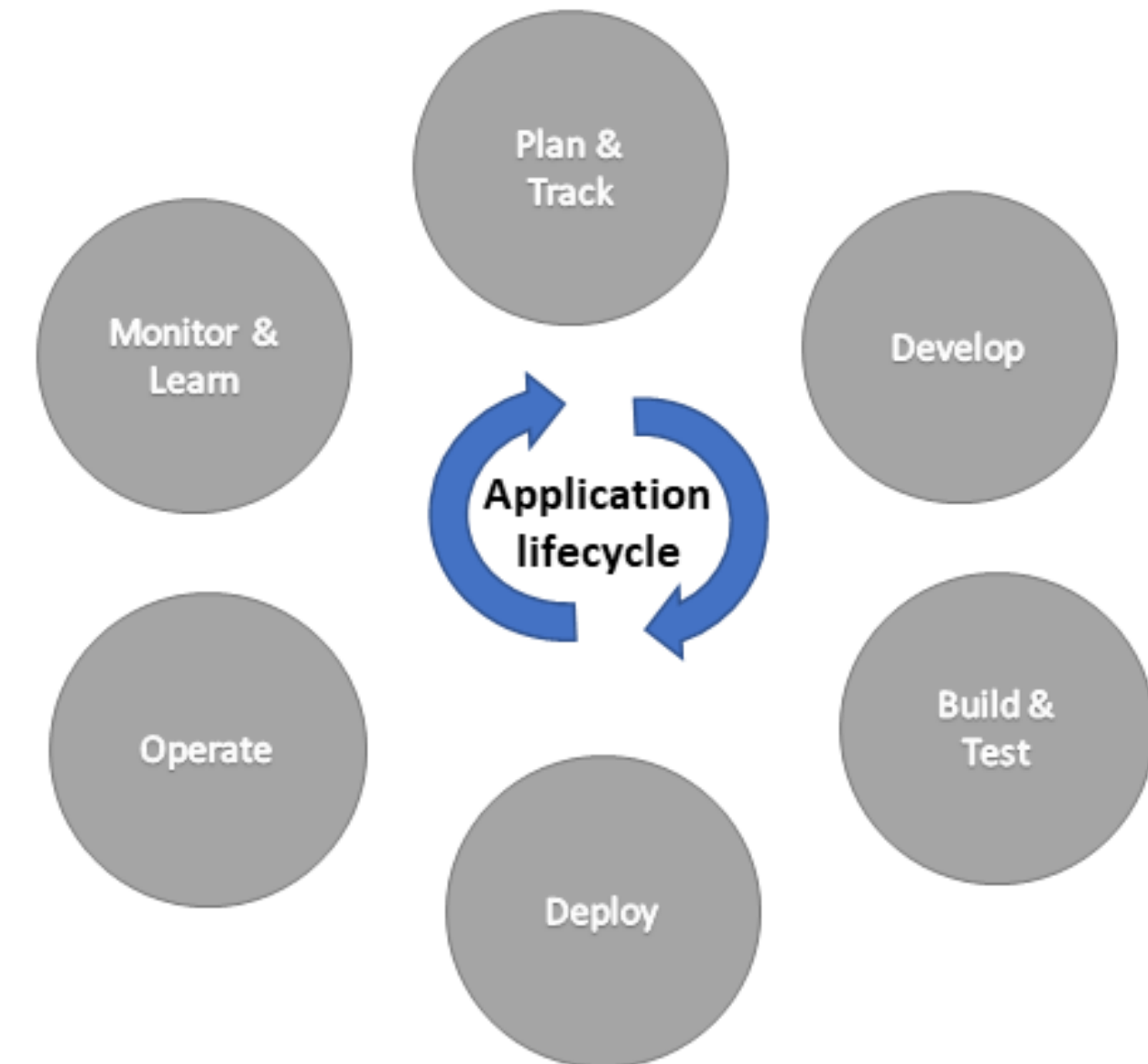
Module 4: Going Further - Advanced Concepts and Best Practices



Friday deployment – what could go wrong? 🤔

What is Application Lifecycle Management (ALM)?

- The processes and tools to help you deploy apps, automations and other components:
 - Continuously
 - Safely
 - In a scalable fashion (for 5 or 5,000+ users)
 - Automated (optional)
- Helps make your solution fire, rather than putting production on fire 🙈 🔥



There's a lot to think about...

Key concepts

- Source (version) control
 - Git repositories
- Solutions / Publishers
 - Managed versus unmanaged
- Environments
- Power Platform pipelines



Now you realise you should have gone to the ALM workshop

Power Platform pipelines

The screenshot shows the Power Apps Pipelines interface. The top navigation bar includes the Power Apps logo, a search bar, and the current environment 'Pipeline Dev'. The left sidebar shows the navigation menu with 'Pipelines' selected. The main content area displays the 'ABC Pipeline' details. The 'Development' stage is highlighted with a red box, showing the solution version '1.0.0.0'. The 'Test stage' is currently deploying version '1.0.0.1'. The 'Production stage' is ready for deployment. Arrows indicate the flow from Development to Test stage and then to Production stage. Each stage has a 'Go to this environment' link and a 'Deploy here' button.

Power Apps Search Environment Pipeline Dev

Pipelines Preview

As your app advances, you'll start sharing it to be used in your organization. As it gets widely adopted, it's important to practice healthy application lifecycle management by building securely in an isolated location so the people using your app don't run into issues while you work on it. Once everything is ready, you'll move it to a more permanent place and share there. [Learn more](#)

Pipeline: ABC Pipeline

Details Run history

Development

Securely test and verify in isolation. After you've got everything working properly, it's time to deploy. [Learn more](#)

Solution version: 1.0.0.0

Test stage Go to this environment

Pipelines Test

Currently deploying version 1.0.0.1

Production stage Go to this environment

Pipelines Prod

Deploy here



Time for the weigh up

Power Platform pipelines

- **Advantages**

- No external tooling required
- Supports approval based delegated deployments
- Multi-stage deployment processes
- Generate deployment notes automatically using Copilot
- Deploy immediately or based on a schedule.

- **Disadvantages**

- Requires Managed Environments; all users accessing the environment must be licensed.
- Not suitable for complex deployments:
 - Multiple solutions
 - Azure components
 - Configuration data



Demo: Power Platform pipelines



It's a tough job, but someone's got to do it

Testing, Debugging and Troubleshooting your apps

- Unit test your work!
 - Manually
 - Use Test Studio
- Next, perform User Acceptance Testing (UAT)
 - Configure a dedicated environment for this
- Use **Trace()** function in canvas apps and custom pages for additional debugging capability
- Once live:
 - Review analytic reports regularly
 - User having an issue? Jump onto Monitor with them!



Gotta have more studios

Test studio

Power Apps | Test Studio - Onboarding Tasks

Environment Development

Save Publish Play Copy play link

Tests

New suite

Test

Suite

Case

Suite

Enter a suite description

Result steps not applied View

Case

Enter a case description

Order	Step	Screen	Action	Play status
1.	Enter a step description	-	Enter a step action	-

Provide feedback



Remote debugging FTW

Power Apps Monitor

Monitor is a tool that you can launch from Microsoft Power Apps Studio to help you troubleshoot problems and improve the quality of your apps

Microsoft Power Apps | Monitor - Bikes

Environment: Contoso

Clear data Upload Download Invite Connect user Play published app Default Filter

Published app session (See versions)

I..	Time	Category	Operation	Result	R...	Status	Duration (...)	Data source	Control	Property	Response ...
16	17:43:18.904	ScreenLoad	Navigate (Brow...	Success			150		IconNewItem1	OnSelect	
17	17:43:19.686	UserAction	Select	Success					DataCardValue2	OnSelect	
18	17:43:30.923	UserAction	SetProperty	Success					DataCardValue2	Text	
19	17:43:30.958	UserAction	Select	Success					IconAccept1	OnSelect	
20	17:43:31.264	Network	createRow	Success	Created	201	247	Bikes	IconAccept1	OnSelect	1,493
21	17:43:31.452	ScreenLoad	Navigate (Brow...	Success			155		EditForm1	OnSuccess	
22	17:43:31.471	Network	getRows	Success		200	123	Bikes	BrowseGallery1	Items	3,008
23	17:43:31.603	Network	getAttachments	Success		200	230	Bikes	Attachments_D...	Default	122
24	17:43:32.390	UserAction	Select	Success					IconNewItem1	OnSelect	
25	17:43:32.553	ScreenLoad	Navigate (Brow...	Success			150		IconNewItem1	OnSelect	
26	17:43:33.887	UserAction	Select	Success					DataCardValue2	OnSelect	
27	17:43:36.620	UserAction	SetProperty	Success					DataCardValue2	Text	
28	17:43:36.670	UserAction	Select	Success					IconAccept1	OnSelect	
29	17:43:36.949	Network	createRow	Success	Created	201	238	Bikes	IconAccept1	OnSelect	1,499
30	17:43:37.121	ScreenLoad	Navigate (Brow...	Success			151		EditForm1	OnSuccess	
31	17:43:37.130	Network	getRows	Success		200	127	Bikes	BrowseGallery1	Items	4,508
32	17:43:37.180	Network	getAttachments	Success		200	204	Bikes	Attachments_D...	Default	122

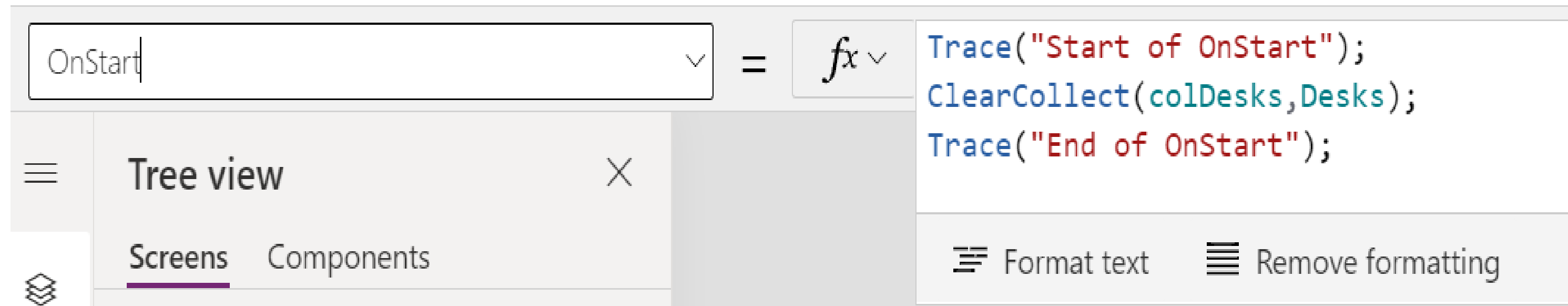
Connected Items: 32 Provide feedback



Console Write Log is soooooo 1996 🙋

Trace

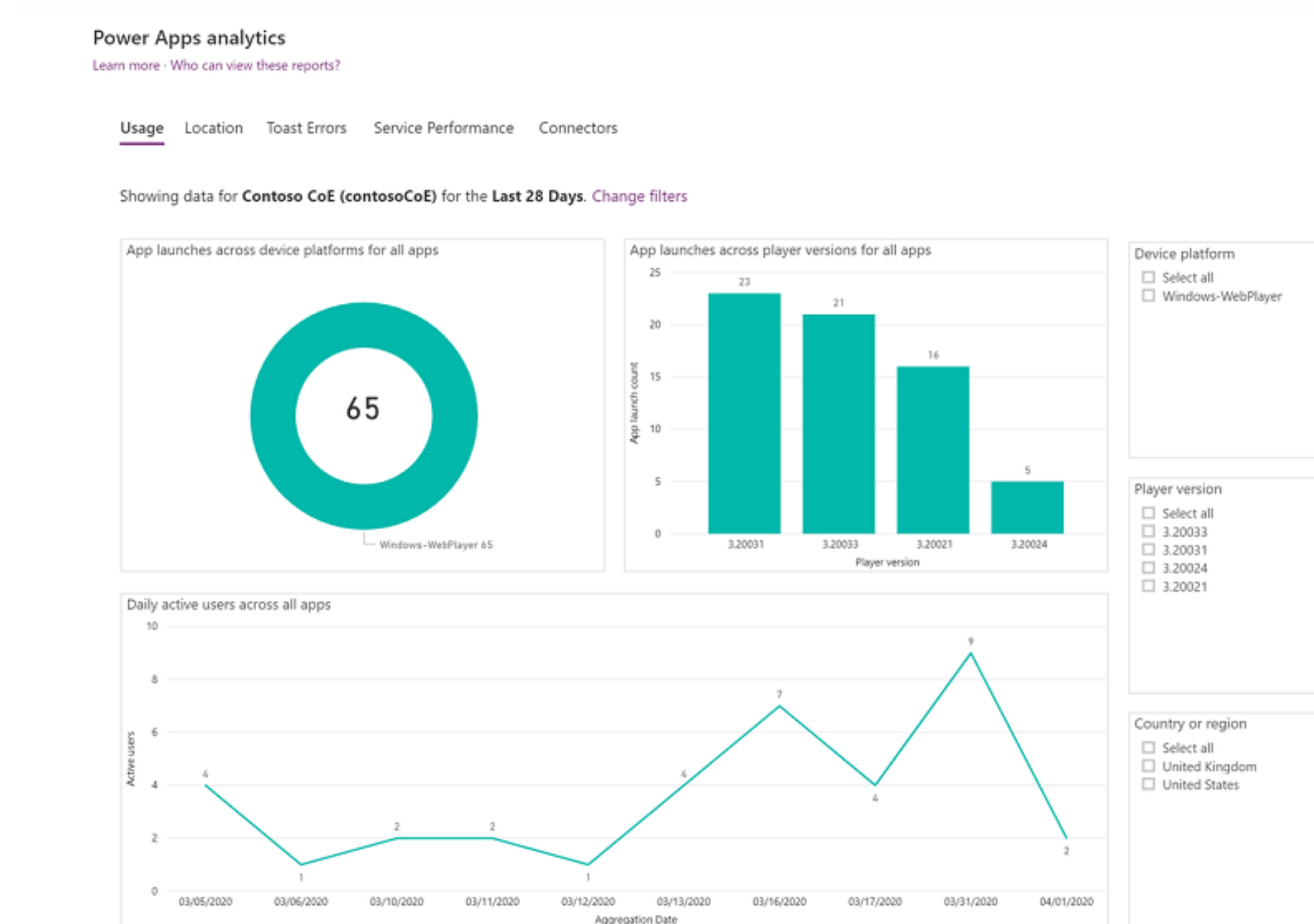
Add custom trace messages for Monitor



Gotta close that feedback loop

Analytics

Power Apps analytics are available via the Power platform admin center: <https://aka.ms/ppac>



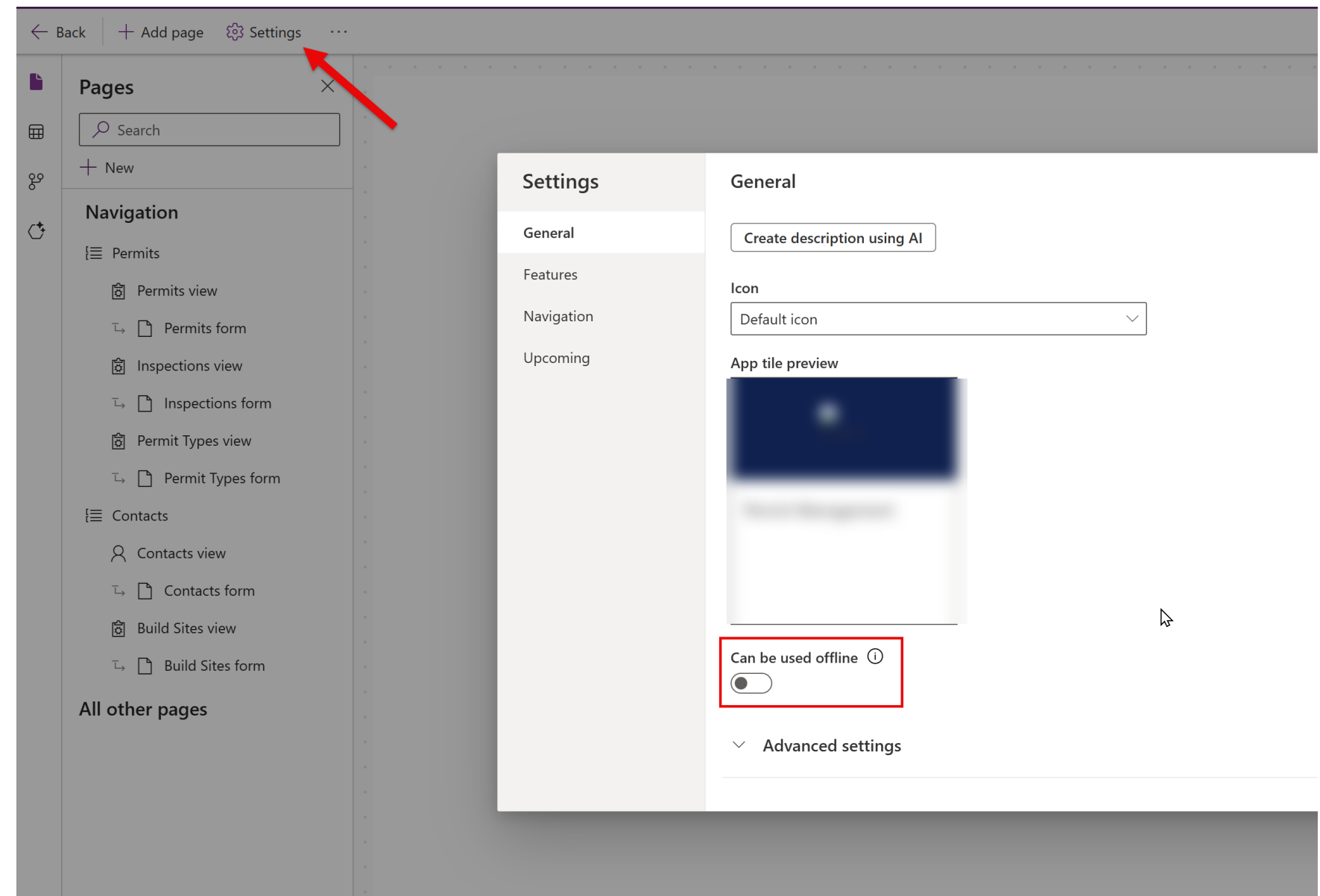
Demo: Test Studio, Monitor and Analytic Reports



Digital detox for your apps 🧘

Offline app considerations

- Custom pages / canvas apps:
 - Use **LoadData** and **SaveData** functions:
 - Consider device-level storage limitations (30-70MB)
 - Merge conflicts - how to handle?
- Model-driven apps
 - Each app must be enabled for offline use
 - Configure offline profile to define the table / data to store locally
 - Each table must be enabled for offline first



Thought you could escape this, huh? 😊

Copilot feature overview

- For canvas apps and custom pages:
 - Create and explain Power Fx formulas
 - Copilot chat



Write formulas like a boss 🦊

Create and explain Power Fx formulas

```
//Check what day of the week it is - if Monday to Friday, use Color.White. Otherwise, use Color.Black
```

< 1/1 > Accept Make sure AI-generated content is accurate and appropriate before using. [See preview terms](#) ...

Otherwise, use Color.Black

```
If(Weekday(Today()) in [ 2, 3, 4, 5, 6 ], Color.White, Color.Black)
```

- Use Copilot to:
 - Explain a complex formula for you
 - Suggest a formula based on comments.
- Included as part of your licensing; no additional purchase necessary.



Help me, chat!

Copilot chat

- Built-in assistant, directly within your app.
- Key benefits:
 - Ask and answer questions regarding Dataverse data...
 - ...or get some hints on potential questions you can ask
 - Quickly open records using the record picker
- Generally available for Dynamics 365 apps, public preview for custom Power Apps

The screenshot displays the 'Innovation Challenge' app in Power Apps. The main view is a table titled 'Active Challenges' with columns for Name, Number of..., Launch date, Accept new ide..., Close Date, and Awards issued?. The table lists various challenges like 'Connected Operations', 'Enterprise sustainability', and '3D Printing'. On the right, the Copilot chat interface is open, showing suggestions for actions and analyses. A red box highlights the 'Navigate to challenges' button and the confirmation message 'We navigated you to the Challenges page'.

Name	Number of...	Launch date	Accept new ide...	Close Date	Awards issued?
Connected Operations	5	3/12/2018	4/29/2018	5/5/2018	No
Enterprise sustainability	4	4/16/2018	4/30/2018	5/3/2018	No
Connected products	3	10/1/2018	10/30/2018	11/4/2018	No
3D Printing	2	3/1/2018	4/30/2018	5/3/2018	No
Smarter manufacturing	2	5/15/2018	5/29/2018	6/2/2018	No
Big data	2	12/1/2018	12/30/2018	1/4/2019	No
Servitization	1	8/1/2018	8/29/2018	9/2/2018	No
Renewable energy	1	9/1/2018	9/30/2018	10/4/2018	No
Holographic computing	0	7/1/2018	7/29/2018	8/2/2018	No



Got some links for you

Additional resources – Custom Page design

- Bootstrap:
 - <https://icons.getbootstrap.com/>
- CSS blur glass morphism:
 - <https://css.glass/>





Lab 4 – DEPLOY AND TEST THE SOLUTION

In this lab, you will do the following:

- Export the solution as a managed solution from the Development environment.
- Import the managed solution into the Production environment.
- Test the solution in the Production environment.

Expected completion time: 30 minutes



Closing thoughts

What's next for your learning experience?

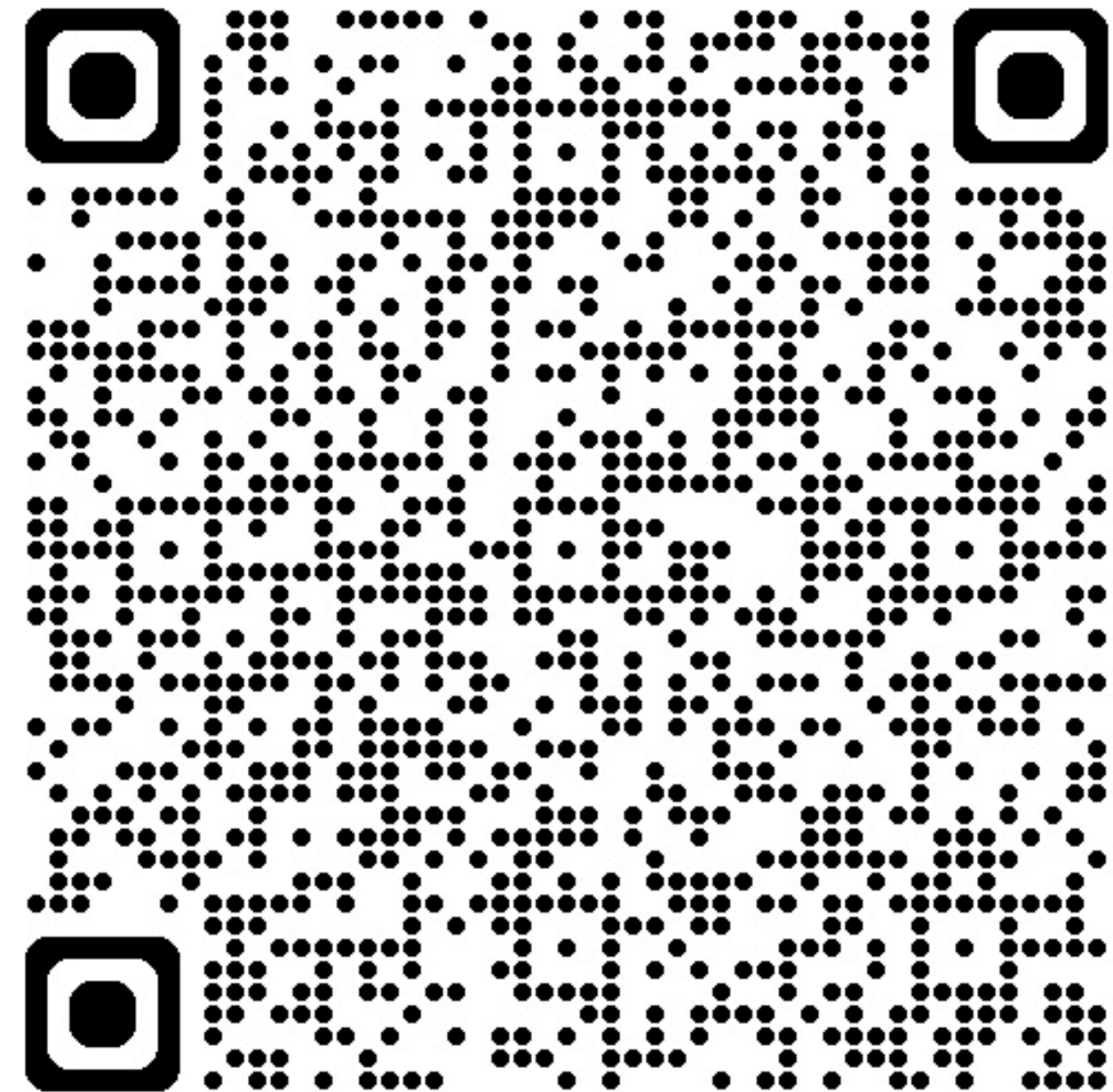
- Liking to use something goes a LONG way to helping with user adoption 🤝
- Convergence of canvas versus model-driven gradually closing; transformative potential for all your user interfaces 💡
- Build solutions that utilises components of Power Platform in the right way
- Low code doesn't necessarily mean low complexity; strike a balance ⚖️
- Use generative AI to your advantage... 🤖
- ...but a bit of knowledge certainly helps when vibe coding 😊
- Go play around with vibe.powerapps.com, Code Apps and generative pages!



How did we do today? 🥹

Your feedback is greatly appreciated!

- Please submit your feedback on today's workshop:
 - What was good 👍
 - What was bad 👎
 - How you rate our (obviously) hilarious slide titles 😂



SESSION ASSETS & SURVEY

<https://linktr.ee/canadiansummit>

View on mobile





THANK YOU!

